# Set 37: Direct proof with numbers

Local outcome LOGIC.PROOF.01: Build a direct argument about even, odd and multiples from definitions

Formal logic set · genre: formal logic · MP3, MP7

- MAIN: Build a direct argument about even, odd and multiples from definitions
- BACKWARD 1: Use the definitions of even and odd
- FORWARD 1: ADVANCED APPLICATION: Direct proofs about multiples
- FORWARD 2: EXPERT CHALLENGE: Judge and complete direct proofs

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Put the steps of a direct proof that the sum of two odd integers is even in order.

Put these in order:

**A** Let m = 2a + 1 and n = 2b + 1, where a and b are integers.

**B** Then m + n = 2a + 2b + 2.

**C** So m + n = 2(a + b + 1).

**D** Since a + b + 1 is an integer, m + n is even.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-M-Q1**
Logic, Set 37: Direct proof with numbers · MAIN · Q1/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Which first line correctly begins a direct proof that the product of any two odd integers is odd?

**A.** Let m = 2a + 1 and n = 2a + 1, where a is an integer.

**B.** Let m = 3 and n = 5.

**C.** Let m = 2a + 1 and n = 2b + 1, where a and b are integers.

**D.** Assume that m times n is odd.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-M-Q2**
Logic, Set 37: Direct proof with numbers · MAIN · Q2/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. A proof that an even integer plus an odd integer is odd reaches the line m + n = 2a + 2b + 1. Which line correctly finishes it?

**A.** Since 4 + 3 = 7 is odd, m + n is odd.

**B.** m + n is odd because the expression 2a + 2b + 1 contains a 1.

**C.** m + n = 2(a + b + 1), so m + n is even.

**D.** m + n = 2(a + b) + 1, and a + b is an integer, so m + n is odd.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-M-Q3**
Logic, Set 37: Direct proof with numbers · MAIN · Q3/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Checking that 3 + 5, 7 + 9 and 11 + 13 are all even proves that the sum of any two odd integers is even.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-M-Q4**
Logic, Set 37: Direct proof with numbers · MAIN · Q4/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. To show that 6n + 4 is even for every integer n, write 6n + 4 = 2(pn + q). Give the numbers p and q.

p = _____

q = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-M-Q5**
Logic, Set 37: Direct proof with numbers · MAIN · Q5/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Which expression is odd for every integer k?

**A.** 2k + 7

**B.** k + 1

**C.** 2k + 4

**D.** 4k

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S37-B1-Q1**
Logic, Set 37: Direct proof with numbers · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. 2k + 6 is even for every integer k.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-B1-Q2**
Logic, Set 37: Direct proof with numbers · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Write 19 in the form 2k + 1. What is k?

k = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S37-B1-Q3**
Logic, Set 37: Direct proof with numbers · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. If n = 2k + 1, then n + 1 is odd.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-B1-Q4**
Logic, Set 37: Direct proof with numbers · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Which statement is the definition of an even integer?

**A.** n ends in the digit 0

**B.** n = 2k + 1 for some integer k

**C.** n leaves remainder 1 when divided by 2

**D.** n = 2k for some integer k

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S37-B1-Q5**
Logic, Set 37: Direct proof with numbers · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Put the steps of a direct proof that if n is a multiple of 3, then n squared is a multiple of 9 in a correct order. (Two orders are correct.)

Put these in order:

**A** Let n = 3k, where k is an integer.

**B** Then n squared = 9k squared.

**C** k squared is an integer, because k is an integer.

**D** So n squared is 9 times an integer, a multiple of 9.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-F1-Q1**
Logic, Set 37: Direct proof with numbers · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Which is a correct direct proof that the sum of three consecutive integers is a multiple of 3?

**A.** n + (n + 1) + (n + 2) = 3n + 3 = 3(n + 1), and n + 1 is an integer.

**B.** 1 + 2 + 3 = 6, and 6 is a multiple of 3.

**C.** n + (n + 1) + (n + 2) = 3n + 3, which is odd.

**D.** n + n + n = 3n, which is a multiple of 3.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-F1-Q2**
Logic, Set 37: Direct proof with numbers · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. If n is odd, then n squared minus 1 is a multiple of 8.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-F1-Q3**
Logic, Set 37: Direct proof with numbers · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. (2a + 1)(2b + 1) = 2(2ab + a + b) + 1. For a = 3 and b = 4, what number is 2ab + a + b?

Number = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-F1-Q4**
Logic, Set 37: Direct proof with numbers · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. A student proves that every multiple of 4 is even by writing n = 4k = 2(2k). What justifies the last sentence, "so n is even"?

**A.** n can be any number at all.

**B.** 2k is an integer, so n is 2 times an integer.

**C.** 4 is an even number.

**D.** k must be an even number.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-F1-Q5**
Logic, Set 37: Direct proof with numbers · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Put the steps of a direct proof that the square of an odd integer leaves remainder 1 when divided by 4 in order.

Put these in order:

**A** Let n = 2k + 1, where k is an integer.

**B** Then n squared = 4k squared + 4k + 1.

**C** So n squared = 4(k squared + k) + 1.

**D** Since k squared + k is an integer, n squared leaves remainder 1 when divided by 4.

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S37-F2-Q1**
Logic, Set 37: Direct proof with numbers · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Claim: the sum of two even integers is a multiple of 4. "Proof: Let m = 2k and n = 2k. Then m + n = 4k." What is wrong?

**A.** Nothing is wrong; the proof is correct.

**B.** It should have started with m = 4k.

**C.** It uses the same k for both, so it only covers the case m = n.

**D.** It should check an example before starting.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-F2-Q2**
Logic, Set 37: Direct proof with numbers · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. The statement "n squared + n is even for every integer n" can be proved by two cases: n even and n odd.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-F2-Q3**
Logic, Set 37: Direct proof with numbers · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. For odd n = 2k + 1, n squared + 4n + 3 = 2(2k squared + 6k + 4). For k = 5, what number is 2k squared + 6k + 4?

Number: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S37-F2-Q4**
Logic, Set 37: Direct proof with numbers · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Give an integer n from 1 to 20 for which n squared + n + 1 is even. If no such n exists, enter 0.

n = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S37-F2-Q5**
Logic, Set 37: Direct proof with numbers · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.PROOF.01 (local) · MP3, MP7

# Set 38: Tiling and covering

Local outcome LOGIC.SPATIAL.05: Decide which pieces cover a region exactly, and prove when no covering exists

Puzzle set · genre: tiling · MP1, MP7

MAIN: Decide which pieces cover a region exactly, and prove when no covering exists

BACKWARD 1: Strategy: color the board like a checkerboard

FORWARD 1: ADVANCED APPLICATION: Count tilings and use other pieces

FORWARD 2: EXPERT CHALLENGE: Prove impossibility and count harder tilings

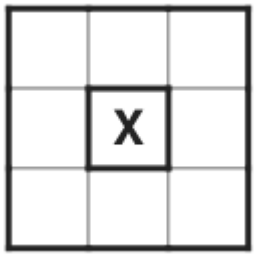
A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. In how many different ways can a rectangle 2 squares tall and 4 squares wide be tiled with dominoes?



Ways: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S38-M-Q1**
Logic, Set 38: Tiling and covering · MAIN · Q1/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. The 3 by 3 board with its center square removed (shown) can be tiled by 4 dominoes.

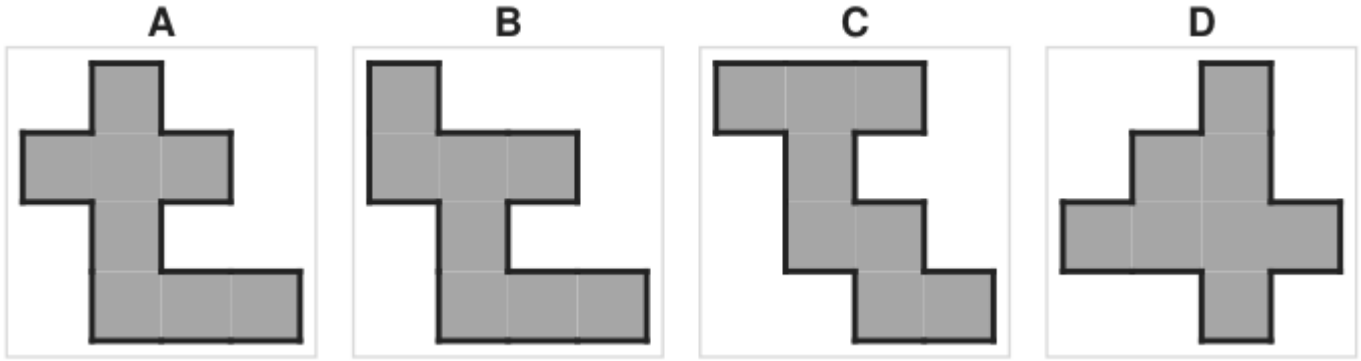



**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-M-Q2**
Logic, Set 38: Tiling and covering · MAIN · Q2/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. Which region can be tiled by 4 dominoes?




**A.** Figure A
**B.** Figure B
**C.** Figure C
**D.** Figure D

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S38-M-Q3**
Logic, Set 38: Tiling and covering · MAIN · Q3/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. One square is removed from a 5 by 5 board, and the other 24 squares are to be tiled by 12 dominoes. Rows and columns are numbered 1 to 5. Give the row r and column c of a square whose removal makes a tiling possible.

Row r = _____

Column c = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-M-Q4**
Logic, Set 38: Tiling and covering · MAIN · Q4/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

An L-tromino is three squares in an L shape (a 2 by 2 square with one square missing). It may be turned any way. In how many ways can a rectangle 2 squares tall and 3 wide be tiled by two L-trominoes?

Ways: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S38-M-Q5**
Logic, Set 38: Tiling and covering · MAIN · Q5/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A 4 by 4 board is colored like a checkerboard. How many of its squares are dark?

Dark: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S38-B1-Q1**
Logic, Set 38: Tiling and covering · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

Every domino placed on a checkerboard covers one dark square and one light square.

**A.** True
**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-B1-Q2**
Logic, Set 38: Tiling and covering · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. A region has 7 dark and 5 light squares on a checkerboard coloring. According to the color count, what is the greatest number of dominoes that could be placed on it without overlapping?

Dominoes: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S38-B1-Q3**
Logic, Set 38: Tiling and covering · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. A region has 6 dark and 6 light squares. What does the color count tell you?

**A.** It can be tiled in exactly 6 ways.

**B.** It can certainly be tiled by 6 dominoes.

**C.** It does not rule out a tiling; you still have to find one or show none exists.

**D.** It cannot be tiled by dominoes.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-B1-Q4**
Logic, Set 38: Tiling and covering · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. A region with an even number of squares can always be tiled by dominoes.

**A.** True
**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-B1-Q5**
Logic, Set 38: Tiling and covering · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. In how many ways can a strip 2 squares tall and 6 squares wide be tiled by dominoes?

Ways: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S38-F1-Q1**
Logic, Set 38: Tiling and covering · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A straight tromino covers three squares in a row. A 3 by 3 square can be tiled by three straight trominoes in exactly 2 ways.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-F1-Q2**
Logic, Set 38: Tiling and covering · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A straight tromino covers three squares in a row, flat or upright. One corner square is removed from a 4 by 4 board. Can the other 15 squares be tiled by 5 straight trominoes?

**A.** Yes, and there is more than one way.

**B.** No: the corner gap blocks every tiling.

**C.** No: 15 squares cannot be split into trominoes.

**D.** Yes, but in exactly one way.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-F1-Q3**
Logic, Set 38: Tiling and covering · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A straight tromino covers three squares in a row, flat or upright. In how many ways can a rectangle 3 squares tall and 4 wide be tiled by straight trominoes?

Ways: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S38-F1-Q4**
Logic, Set 38: Tiling and covering · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. A strip 2 squares tall and n squares wide can be tiled by dominoes in exactly 8 ways. What is n?

n = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S38-F1-Q5**
Logic, Set 38: Tiling and covering · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. In how many ways can a rectangle 3 squares tall and 4 wide be tiled by dominoes?

Ways: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S38-F2-Q1**
Logic, Set 38: Tiling and covering · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. A board 4 rows by 5 columns has two squares removed: row 1, column 1 and row 3, column 3. The other 18 squares can be tiled by 9 dominoes.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S38-F2-Q2**
Logic, Set 38: Tiling and covering · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

An L-tromino is three squares in an L shape, turned any way. In how many ways can a strip 2 squares tall and 9 wide be tiled by L-trominoes?

**A.** 6
**B.** 8
**C.** 9
**D.** 3

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S38-F2-Q3**
Logic, Set 38: Tiling and covering · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. One square is removed from a 3 by 3 board. Rows and columns are numbered 1 to 3. Give a square (row r, column c) whose removal leaves 8 squares that can be tiled by dominoes.

Row r = _____

Column c = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S38-F2-Q4**
Logic, Set 38: Tiling and covering · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. One square is removed from a board 3 rows by 5 columns. For how many of the 15 squares does removing it leave 14 squares that can be tiled by dominoes?

Squares: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S38-F2-Q5**
Logic, Set 38: Tiling and covering · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.SPATIAL.05 (local) · MP1, MP7

# Set 39: Justify each step of solving an equation

Local outcome LOGIC.PROOF.02: Explain each step in solving an equation and justify the solution method

Formal logic set · genre: formal logic · MP3, MP6

MAIN: Explain each step in solving an equation and justify the solution method
BACKWARD 1: Properties of equality
FORWARD 1: ADVANCED APPLICATION: Steps that need care
FORWARD 2: EXPERT CHALLENGE: Justify a solution method

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Put the steps for solving $4x + 7 = 2x + 19$ in order.

Put these in order:

**A** $2x + 7 = 19$ (subtract $2x$ from both sides)

**B** $2x = 12$ (subtract 7 from both sides)

**C** $x = 6$ (divide both sides by 2)

**D** Check: $4(6) + 7 = 31$ and $2(6) + 19 = 31$

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-M-Q1**
Logic, Set 39: Justify each step of solving an equation · MAIN · Q1/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Which reason justifies going from 5x - 4 = 21 to 5x = 25?

**A.** Use the distributive property on 5x - 4.

**B.** Multiply both sides by 4.

**C.** Rewrite 5x - 4 as 4 - 5x.

**D.** Add 4 to both sides; equal numbers plus 4 are still equal.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-M-Q2**
Logic, Set 39: Justify each step of solving an equation · MAIN · Q2/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. A student writes: Line 1: 2(x + 3) = 14. Line 2: 2x + 3 = 14. Line 3: 2x = 11. Line 4: x = 5.5. Which line is the first one that does NOT follow from the line before it?

**A.** Line 2

**B.** None; every line follows

**C.** Line 4

**D.** Line 3

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-M-Q3**
Logic, Set 39: Justify each step of solving an equation · MAIN · Q3/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. From 6 - x = 10, the next line "x = 10 - 6" follows correctly.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-M-Q4**
Logic, Set 39: Justify each step of solving an equation · MAIN · Q4/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Solve 7 - 2(x - 1) = x - 3. Give x.

x = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S39-M-Q5**
Logic, Set 39: Justify each step of solving an equation · MAIN · Q5/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

If a = b, which statement must also be true?

**A.** a / 5 = b / 3

**B.** 5a = b

**C.** a - 5 = b - 5

**D.** a - 5 = b + 5

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-B1-Q1**
Logic, Set 39: Justify each step of solving an equation · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

If $3x = 12$, then $x = 4$ follows by dividing both sides by 3.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-B1-Q2**
Logic, Set 39: Justify each step of solving an equation · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Which step uses the distributive property?

**A.** $4x = 20$ becomes $x = 5$

**B.** $5(x - 2) = 13$ becomes $5x - 10 = 13$

**C.** $x + 3 = 8$ becomes $x = 5$

**D.** $x + 2 = 9$ becomes $2 + x = 9$

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-B1-Q3**
Logic, Set 39: Justify each step of solving an equation · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Multiplying both sides of an equation by 0 always gives an equation with the same solutions.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-B1-Q4**
Logic, Set 39: Justify each step of solving an equation · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

To solve x / 4 = 9 in one step, by what number do you multiply both sides?

Number: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S39-B1-Q5**
Logic, Set 39: Justify each step of solving an equation · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. A student solves x squared = 5x by dividing both sides by x and gets x = 5. What is wrong?

**A.** Nothing; x = 5 is the only solution.

**B.** The answer should be x = 0 only.

**C.** x squared = 5x has no solutions.

**D.** Dividing by x assumes x is not 0, and x = 0 is also a solution.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-F1-Q1**
Logic, Set 39: Justify each step of solving an equation · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

The equation $3(x + 2) = 3x + 6$ is true for every number $x$.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-F1-Q2**
Logic, Set 39: Justify each step of solving an equation · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Solve 2(x - 4) + 3 = 5x + 7. What is x?

x = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S39-F1-Q3**
Logic, Set 39: Justify each step of solving an equation · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Solving $4x + 1 = 4x - 3$ leads to the line $1 = -3$. What does this show?

**A.** The solution is $x = -3$.

**B.** Every number is a solution.

**C.** The equation has no solution.

**D.** The solution is $x = 0$.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-F1-Q4**
Logic, Set 39: Justify each step of solving an equation · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Put the lines of this solution of (x + 5) / 3 = 4 in order.

Put these in order:

**A** Assume x is a number with (x + 5) / 3 = 4.

**B** x + 5 = 12 (multiply both sides by 3)

**C** x = 7 (subtract 5 from both sides)

**D** Check: (7 + 5) / 3 = 4

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-F1-Q5**
Logic, Set 39: Justify each step of solving an equation · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. To solve the equation (square root of (x + 2)) = x, a student squares both sides: x + 2 = x squared, so x = 2 or x = -1. Which answer is correct?

**A.** Only x = 2; x = -1 gives square root of 1 = 1, not -1.

**B.** Neither one.

**C.** Only x = -1.

**D.** Both x = 2 and x = -1.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-F2-Q1**
Logic, Set 39: Justify each step of solving an equation · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. If each line of a solution follows from the line before it, then every number found at the end must solve the original equation.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-F2-Q2**
Logic, Set 39: Justify each step of solving an equation · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it.
Solve 3(2x - 1) = 4(x + 2) - 1. Give x.

x = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S39-F2-Q3**
Logic, Set 39: Justify each step of solving an equation · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Which step can give an equation with MORE solutions than the one before it?

**A.** Dividing both sides by 2
**B.** Adding 7 to both sides
**C.** Multiplying both sides by -3
**D.** Squaring both sides

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-F2-Q4**
Logic, Set 39: Justify each step of solving an equation · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Put the lines of this solution of 5 - 3(x - 1) = 2x - 2 in order.

Put these in order:

**A** 5 - 3x + 3 = 2x - 2 (distributive property)

**B** 8 - 3x = 2x - 2 (combine 5 and 3)

**C** 10 = 5x (add 3x and 2 to both sides)

**D** x = 2 (divide both sides by 5)

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S39-F2-Q5**
Logic, Set 39: Justify each step of solving an equation · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.PROOF.02 (local) · MP3, MP6

# Set 40: Proof by contradiction

Local outcome LOGIC.PROOF.03: Set up and complete an argument by contradiction

Formal logic set · genre: formal logic · MP3

MAIN: Set up and complete an argument by contradiction
BACKWARD 1: Strategy: assume the opposite and look for the impossible
FORWARD 1: ADVANCED APPLICATION: Contradiction with divisibility
FORWARD 2: EXPERT CHALLENGE: Contradiction in larger arguments

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. To prove "there is no largest even integer" by contradiction, how should the proof begin?

**A.** Show that 100 is an even integer.
**B.** Assume there is a largest even integer; call it N.
**C.** Assume there is no largest even integer.
**D.** Assume every even integer is less than 100.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-M-Q1**
Logic, Set 40: Proof by contradiction · MAIN · Q1/5 · Outcome LOGIC.PROOF.03 (local) · MP3

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Put the steps of a proof that if n squared is even, then n is even, in order.

Put these in order:

**A** Assume n squared is even, but n is odd.

**B** Then n = 2k + 1 for some integer k.

**C** So n squared = 2(2k squared + 2k) + 1, which is odd.

**D** This contradicts n squared being even, so n must be even.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-M-Q2**
Logic, Set 40: Proof by contradiction · MAIN · Q2/5 · Outcome LOGIC.PROOF.03 (local) · MP3

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Reaching a statement that contradicts a known fact shows that the assumption was false.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-M-Q3**
Logic, Set 40: Proof by contradiction · MAIN · Q3/5 · Outcome LOGIC.PROOF.03 (local) · MP3

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Claim: no integer is both even and odd. Suppose n = 2a and n = 2b + 1 for integers a and b. Which line gives the contradiction?

**A.** 2a = 2b + 1, so a = b.

**B.** n = 4 is even and not odd.

**C.** 2a = 2b + 1, so 2(a - b) = 1, but 1 is not 2 times an integer.

**D.** 2a + 2b + 1 is an odd number.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-M-Q4**
Logic, Set 40: Proof by contradiction · MAIN · Q4/5 · Outcome LOGIC.PROOF.03 (local) · MP3

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. To show that 15 is not the sum of two even integers, assume 15 = 2a + 2b for integers a and b. Then a + b must equal what number?

a + b = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-M-Q5**
Logic, Set 40: Proof by contradiction · MAIN · Q5/5 · Outcome LOGIC.PROOF.03 (local) · MP3

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. To prove "n is not divisible by 4" by contradiction, what do you assume?

**A.** n is divisible by 4.

**B.** n is not divisible by 4.

**C.** n is divisible by 2.

**D.** n is an odd number.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-B1-Q1**
Logic, Set 40: Proof by contradiction · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.PROOF.03 (local) · MP3

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. To prove "there is no integer x with 2x = 7" by contradiction, what do you assume?

**A.** Every integer x has 2x = 7.
**B.** There is an integer x with 2x not equal to 7.
**C.** There is no integer x with 2x = 7.
**D.** There is an integer x with 2x = 7.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-B1-Q2**
Logic, Set 40: Proof by contradiction · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.PROOF.03 (local) · MP3

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. A proof by contradiction begins by assuming that the statement to be proved is true.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-B1-Q3**
Logic, Set 40: Proof by contradiction · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.PROOF.03 (local) · MP3

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. If assuming P leads to 1 = 0, then P is false.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-B1-Q4**
Logic, Set 40: Proof by contradiction · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.PROOF.03 (local) · MP3

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Assume there is a largest odd integer M. Which number gives the contradiction?

**A.** 2M

**B.** M + 2

**C.** M - 2

**D.** M + 1

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-B1-Q5**
Logic, Set 40: Proof by contradiction · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.PROOF.03 (local) · MP3

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. To prove that no integers a and b satisfy 6a + 9b = 10, which fact gives the contradiction?

**A.** 6a + 9b = 3(2a + 3b) is a multiple of 3, but 10 is not.

**B.** a and b would have to be fractions.

**C.** 10 is even, and 6a + 9b is always odd.

**D.** 6 and 9 are both odd numbers.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-F1-Q1**
Logic, Set 40: Proof by contradiction · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.PROOF.03 (local) · MP3

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Put the steps of the proof that the square root of 2 is not a fraction in order.

Put these in order:

**A** Assume the square root of 2 = p / q, with p and q integers in lowest terms.
**B** Then p squared = 2q squared, so p squared is even, so p is even: p = 2m.
**C** Then 4m squared = 2q squared, so q squared = 2m squared, and q is even.
**D** Both p and q are even, which contradicts lowest terms.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-F1-Q2**
Logic, Set 40: Proof by contradiction · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.PROOF.03 (local) · MP3

Among any 5 integers, two of them leave the same remainder when divided by 4.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-F1-Q3**
Logic, Set 40: Proof by contradiction · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.PROOF.03 (local) · MP3

A perfect square is divided by 4. What is the largest remainder (0, 1, 2 or 3) that can NEVER occur?

Remainder: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-F1-Q4**
Logic, Set 40: Proof by contradiction · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.PROOF.03 (local) · MP3

Give an integer n from 1 to 50 for which n squared + 1 is a multiple of 3. If no such n exists, enter 0.

n = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S40-F1-Q5**
Logic, Set 40: Proof by contradiction · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.PROOF.03 (local) · MP3

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Assume there are only finitely many primes, p1, p2, ..., pk, and let N = p1 x p2 x ... x pk + 1. Why is this a contradiction?

**A.** N is always an even number.

**B.** N is always a prime number.

**C.** N is larger than every prime, so N is not prime.

**D.** N leaves remainder 1 when divided by each listed prime, so a prime factor of N is not on the list.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-F2-Q1**
Logic, Set 40: Proof by contradiction · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.PROOF.03 (local) · MP3

2 x 3 x 5 x 7 x 11 x 13 + 1 is a prime number.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-F2-Q2**
Logic, Set 40: Proof by contradiction · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.PROOF.03 (local) · MP3

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Put the steps of a proof that if $a + b$ is at least 20, then $a$ is at least 10 or $b$ is at least 10, in order.

Put these in order:

**A** Assume $a + b$ is at least 20, but $a < 10$ and $b < 10$.

**B** Adding the two inequalities gives $a + b < 20$.

**C** That contradicts $a + b$ being at least 20.

**D** So $a$ is at least 10 or $b$ is at least 10.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-F2-Q3**
Logic, Set 40: Proof by contradiction · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.PROOF.03 (local) · MP3

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Claim: every integer is even. "Proof: Assume some integer is odd, say 3. Then 3 = 2k for no integer k. Contradiction." What is wrong?

**A.** Nothing; the proof is correct.

**B.** Nothing impossible was reached; 3 not being 2k agrees with 3 being odd.

**C.** It should assume that every integer is odd.

**D.** It should use 5 instead of 3.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S40-F2-Q4**
Logic, Set 40: Proof by contradiction · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.PROOF.03 (local) · MP3

At least how many integers must you choose to be sure that two of them leave the same remainder when divided by 6?

Integers: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S40-F2-Q5**
Logic, Set 40: Proof by contradiction · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.PROOF.03 (local) · MP3

# Set 41: Parity and alternation

Local outcome LOGIC.INVARIANT.01: Use odd/even and alternation to predict outcomes and rule out possibilities

Puzzle set · genre: parity · MP7, MP8

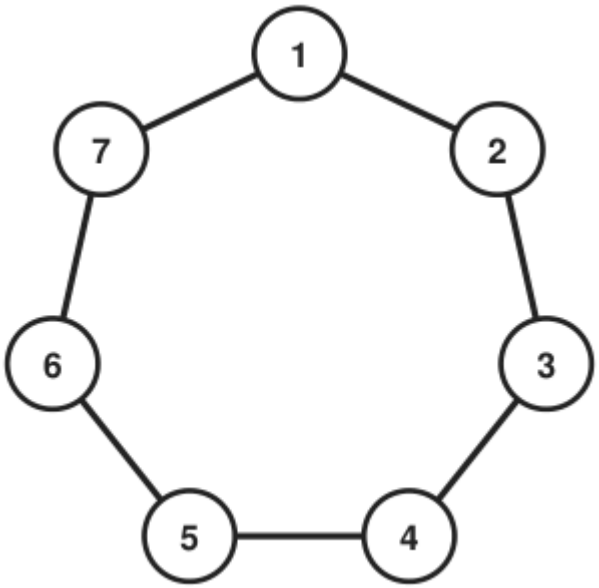
MAIN: Use odd/even and alternation to predict outcomes and rule out possibilities

BACKWARD 1: Strategy: keep track of odd and even only

FORWARD 1: ADVANCED APPLICATION: Parity on boards and rings

FORWARD 2: EXPERT CHALLENGE: Prove impossibility with parity

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Seven gears form a closed ring, each touching the gears on either side (figure). Gear 1 is turned clockwise. What happens?




**A.** Only gears 1, 3, 5 and 7 turn.

**B.** Every gear turns, and the directions alternate around the ring.

**C.** The gears jam: going around the ring, gear 1 would have to turn both ways.

**D.** Every gear turns clockwise.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-M-Q1**
Logic, Set 41: Parity and alternation · MAIN · Q1/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Twelve gears stand in a row (not a ring), each touching the next. Gear 1 turns clockwise. How many of the 12 gears turn counterclockwise?

Gears: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S41-M-Q2**
Logic, Set 41: Parity and alternation · MAIN · Q2/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

A frog starts on 0 on a number line and makes exactly 9 jumps, each 1 unit left or right. It can end on 4.

**A.** True
**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S41-M-Q3**
Logic, Set 41: Parity and alternation · MAIN · Q3/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

A lamp starts OFF. Each press of its switch changes it (OFF to ON, or ON to OFF). Give a number of presses from 1 to 20 that leaves the lamp ON.

Presses = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S41-M-Q4**
Logic, Set 41: Parity and alternation · MAIN · Q4/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

A token moves along a row of squares numbered 1 to 15. Each move jumps exactly 2 squares left or right (staying on the row). It starts on square 4. How many of the 15 squares can it ever reach, counting square 4?

Squares: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S41-M-Q5**
Logic, Set 41: Parity and alternation · MAIN · Q5/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

The sum of 5 odd numbers is odd.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-B1-Q1**
Logic, Set 41: Parity and alternation · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

A coin starts heads up. It is turned over 13 times. Which side is up at the end?

**A.** Either side, depending on how it is turned
**B.** Tails
**C.** Heads
**D.** It cannot be known without trying

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S41-B1-Q2**
Logic, Set 41: Parity and alternation · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Nine gears stand in a row, each touching the next. Gear 1 turns clockwise. What is the number of the last gear in the row that turns clockwise?

Gear: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-B1-Q3**
Logic, Set 41: Parity and alternation · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

An even number plus an odd number is even.

**A.** True
**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S41-B1-Q4**
Logic, Set 41: Parity and alternation · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

Which total CANNOT be the sum of three odd numbers?

**A.** 15
**B.** 9
**C.** 10
**D.** 3

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-B1-Q5**
Logic, Set 41: Parity and alternation · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

A token starts on a corner square of a 5 by 5 board. Each move goes to a square sharing a side (going back is allowed). After exactly 7 moves, how many different squares could the token be on?

Squares: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S41-F1-Q1**
Logic, Set 41: Parity and alternation · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Ten gears form a closed ring, each touching the gears on either side. They can all turn.

**A.** True
**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S41-F1-Q2**
Logic, Set 41: Parity and alternation · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

Seven cups stand upside down. A move turns over exactly 4 cups (any 4). Can all seven cups end right side up?

**A.** No: each move changes the number of upside-down cups by an even amount, so it stays odd.

**B.** Yes, after exactly 7 moves.

**C.** Yes: turn over 4 cups, then 3 cups.

**D.** No: 7 is not a multiple of 4.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-F1-Q3**
Logic, Set 41: Parity and alternation · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

Put a + or - sign in front of each of the numbers 1, 2, 3, ..., 10 and add. Give a total from 0 to 10 that can be made this way.

Total = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-F1-Q4**
Logic, Set 41: Parity and alternation · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Nine gears are arranged in a 3 by 3 square, and each gear touches the gears directly beside, above and below it. A corner gear turns clockwise. How many gears turn clockwise, including the corner gear?

Gears: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S41-F1-Q5**
Logic, Set 41: Parity and alternation · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

A token must visit every square of a 5 by 5 board exactly once, each move going to a square that shares a side. It starts on the square next to a corner (row 1, column 2). Is this possible?

**A.** Yes, but only if it ends on a corner.

**B.** Yes: go along row 1, then snake back and forth.

**C.** No: 25 is not a multiple of 5.

**D.** No: the path must start and end on the corner color, and this square has the other color.

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S41-F2-Q1**
Logic, Set 41: Parity and alternation · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

Put a + or - sign in front of each of the numbers 1, 2, 3, ..., 10 and add. How many different totals can be made?

Totals: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S41-F2-Q2**
Logic, Set 41: Parity and alternation · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

On a square grid, a path that moves one square up, down, left or right at each step, and ends where it started, always has an even number of steps.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S41-F2-Q3**
Logic, Set 41: Parity and alternation · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Give a number of gears n from 3 to 12 for which a closed ring of n gears, each touching its two neighbors, would jam.

n = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S41-F2-Q4**
Logic, Set 41: Parity and alternation · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

A bug walks along the edges of a cube from corner to corner. It starts at one corner and makes exactly 5 moves (going back is allowed). How many of the 8 corners could it end on?

Corners: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S41-F2-Q5**
Logic, Set 41: Parity and alternation · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.INVARIANT.01 (local) · MP7, MP8

# Set 42: Disprove a claim with a counterexample

Local outcome LOGIC.PROOF.04: Find a counterexample that disproves a general mathematical claim

Formal logic set · genre: formal logic · MP3

MAIN: Find a counterexample that disproves a general mathematical claim
BACKWARD 1: What a counterexample is
FORWARD 1: ADVANCED APPLICATION: Counterexamples to algebra rules
FORWARD 2: EXPERT CHALLENGE: Counterexamples that take a search

A counterexample to a claim about every case is one case where the claim is false. Claim: for every whole number n, n squared + n + 11 is prime. Which n is a counterexample?

**A.** 1
**B.** 10
**C.** 3
**D.** 2

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S42-M-Q1**
Logic, Set 42: Disprove a claim with a counterexample · MAIN · Q1/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: for all numbers a and b, if a > b, then a squared > b squared. Give a counterexample.

a = _____

b = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S42-M-Q2**
Logic, Set 42: Disprove a claim with a counterexample · MAIN · Q2/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. The claim "every odd number greater than 1 is prime" is disproved by the number 9.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S42-M-Q3**
Logic, Set 42: Disprove a claim with a counterexample · MAIN · Q3/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: 2 to the power n, plus 1, is prime for every whole number n from 1 to 6. What is the smallest n that disproves it?

n = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S42-M-Q4**
Logic, Set 42: Disprove a claim with a counterexample · MAIN · Q4/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: every multiple of 4 is a multiple of 8. Which number is NOT a counterexample?

**A.** 16
**B.** 4
**C.** 12
**D.** 20

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S42-M-Q5**
Logic, Set 42: Disprove a claim with a counterexample · MAIN · Q5/5 · Outcome LOGIC.PROOF.04 (local) · MP3

What is a counterexample to the claim "every number with property P has property Q"?

**A.** A number with property Q that does not have property P

**B.** A number with neither property

**C.** A number with both properties

**D.** A number with property P that does not have property Q

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-B1-Q1**
Logic, Set 42: Disprove a claim with a counterexample · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.PROOF.04 (local) · MP3

One counterexample is enough to show that a claim about every case is false.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S42-B1-Q2**
Logic, Set 42: Disprove a claim with a counterexample · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.PROOF.04 (local) · MP3

If a claim works for five examples, it is proved true for every case.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-B1-Q3**
Logic, Set 42: Disprove a claim with a counterexample · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Which number is a counterexample to "all prime numbers are odd"?

**A.** 3
**B.** 9
**C.** 2
**D.** 4

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-B1-Q4**
Logic, Set 42: Disprove a claim with a counterexample · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: every whole number divisible by 3 is divisible by 6. Give a counterexample from 1 to 30.

n = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S42-B1-Q5**
Logic, Set 42: Disprove a claim with a counterexample · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: (a + b) squared = a squared + b squared for all numbers a and b. Give whole numbers a and b that disprove it.

a = _____

b = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S42-F1-Q1**
Logic, Set 42: Disprove a claim with a counterexample · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: the square root of (a + b) equals the square root of a plus the square root of b, for all numbers a and b that are 0 or more. Which pair is a counterexample?

**A.** a = 0, b = 4

**B.** a = 9, b = 16

**C.** a = 0, b = 0

**D.** a = 1, b = 0

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S42-F1-Q2**
Logic, Set 42: Disprove a claim with a counterexample · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: n factorial plus 1 is prime for every positive whole number n. (n factorial is 1 x 2 x ... x n.) What is the smallest n that disproves it?

n = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S42-F1-Q3**
Logic, Set 42: Disprove a claim with a counterexample · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. The claim "if n squared is a multiple of 4, then n is a multiple of 4" is disproved by n = 2.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-F1-Q4**
Logic, Set 42: Disprove a claim with a counterexample · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: every rectangle with perimeter 20 has area 25. Which rectangle is a counterexample?

**A.** 2 by 10
**B.** 3 by 8
**C.** 5 by 5
**D.** 4 by 6

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S42-F1-Q5**
Logic, Set 42: Disprove a claim with a counterexample · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: for whole numbers a, b and c, if a divides b x c, then a divides b or a divides c. Give a counterexample with a, b and c each from 2 to 12.

a = _____

b = _____

c = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-F2-Q1**
Logic, Set 42: Disprove a claim with a counterexample · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: 2 to the power n, minus 1, is prime whenever n is prime. The smallest counterexample is n = 11.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-F2-Q2**
Logic, Set 42: Disprove a claim with a counterexample · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.PROOF.04 (local) · MP3

Which claim is FALSE?

**A.** The sum of two even numbers is even.

**B.** If n squared is odd, then n is odd.

**C.** If n is prime, then n + 2 is prime.

**D.** If n is odd, then n squared is odd.

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S42-F2-Q3**
Logic, Set 42: Disprove a claim with a counterexample · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: every whole number from 1 to 30 is a sum of two square numbers (0, 1, 4, 9, 16, 25, ...; a square may be used twice). What is the smallest counterexample?

Number: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S42-F2-Q4**
Logic, Set 42: Disprove a claim with a counterexample · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.PROOF.04 (local) · MP3

A counterexample to a claim about every case is one case where the claim is false. Claim: for every positive whole number n, 2 to the power n is greater than n. Why is n = -1 NOT a valid counterexample?

**A.** The claim is only about positive whole numbers, and -1 is not one.

**B.** Because a counterexample must be a large number.

**C.** Because counterexamples cannot be negative.

**D.** Because 2 to the power -1 is greater than -1.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S42-F2-Q5**
Logic, Set 42: Disprove a claim with a counterexample · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.PROOF.04 (local) · MP3

# Set 43: Proofs about lines and angles

Local outcome LOGIC.PROOF.05: Order and justify the steps of proofs about angles formed by intersecting and parallel lines

Formal logic set · genre: formal logic · MP3

MAIN: Order and justify the steps of proofs about angles formed by intersecting and parallel lines
BACKWARD 1: Name the angle pairs
FORWARD 1: ADVANCED APPLICATION: Perpendicular bisectors and converses
FORWARD 2: EXPERT CHALLENGE: Complete and check angle proofs

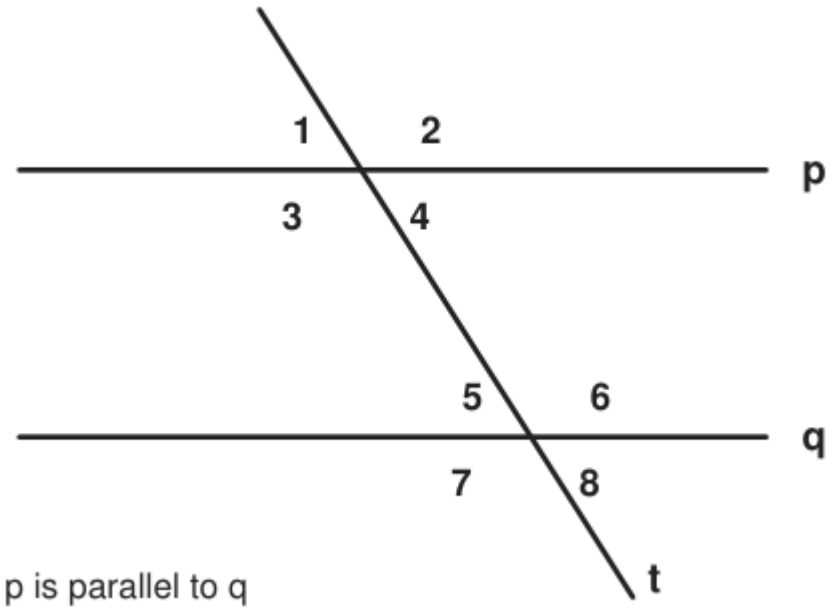
Lines meet at a point, forming angle 1 and angle 2 as vertical angles, and angle 3 next to both (angle 3 forms a linear pair with angle 1 and with angle 2). Put the steps of the proof that angle 1 = angle 2 in a correct order. (Two orders are correct.)

Put these in order:

**A** angle 1 + angle 3 = 180 degrees (linear pair)
**B** angle 2 + angle 3 = 180 degrees (linear pair)
**C** angle 1 + angle 3 = angle 2 + angle 3 (both sums equal 180 degrees)
**D** angle 1 = angle 2 (subtract angle 3 from both sides)

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-M-Q1**
Logic, Set 43: Proofs about lines and angles · MAIN · Q1/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Which pair of angles are alternate interior angles?

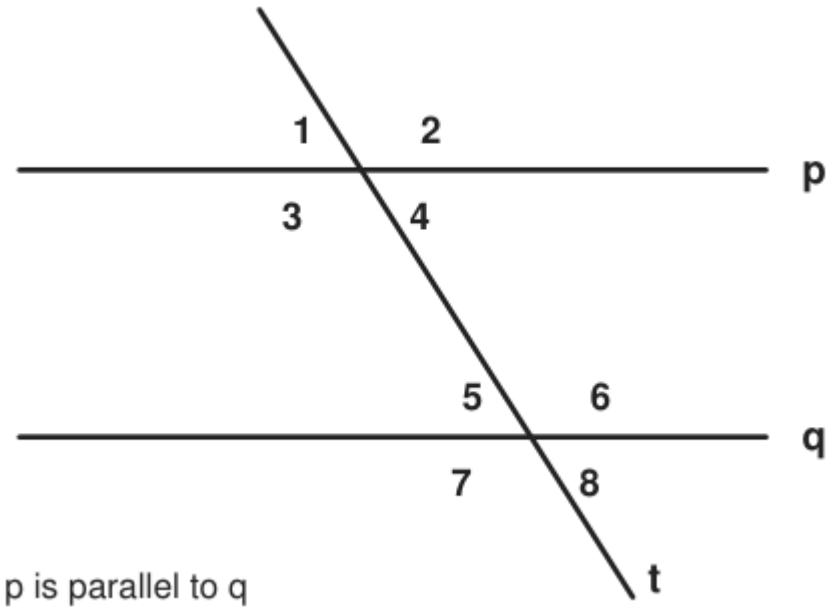



p is parallel to q

**A.** Angles 2 and 3
**B.** Angles 4 and 5
**C.** Angles 4 and 8
**D.** Angles 1 and 5

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-M-Q2**
Logic, Set 43: Proofs about lines and angles · MAIN · Q2/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Lines p and q are parallel. If angle 2 = 70 degrees, then angle 7 = 70 degrees.



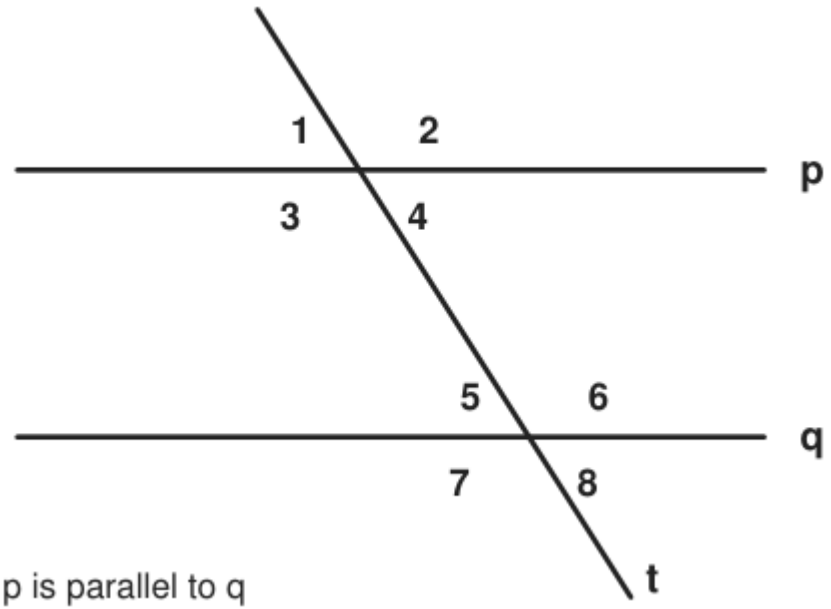

p is parallel to q

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-M-Q3**
Logic, Set 43: Proofs about lines and angles · MAIN · Q3/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Lines p and q are parallel, and angle 1 = 115 degrees. Give the measure of angle 6 in degrees.

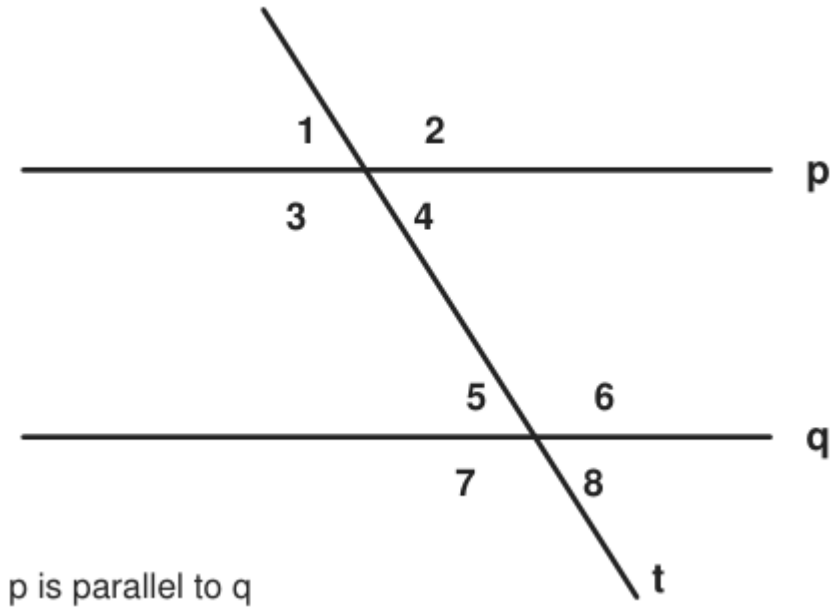



p is parallel to q

angle 6 = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-M-Q4**
Logic, Set 43: Proofs about lines and angles · MAIN · Q4/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Put the steps of the proof that angle 3 = angle 6 when p is parallel to q in a correct order. (Two orders are correct.)




p is parallel to q

Put these in order:

**A** p is parallel to q, cut by transversal t (given)
**B** angle 2 = angle 6 (corresponding angles, p parallel to q)
**C** angle 2 = angle 3 (vertical angles)
**D** angle 3 = angle 6 (both equal angle 2)

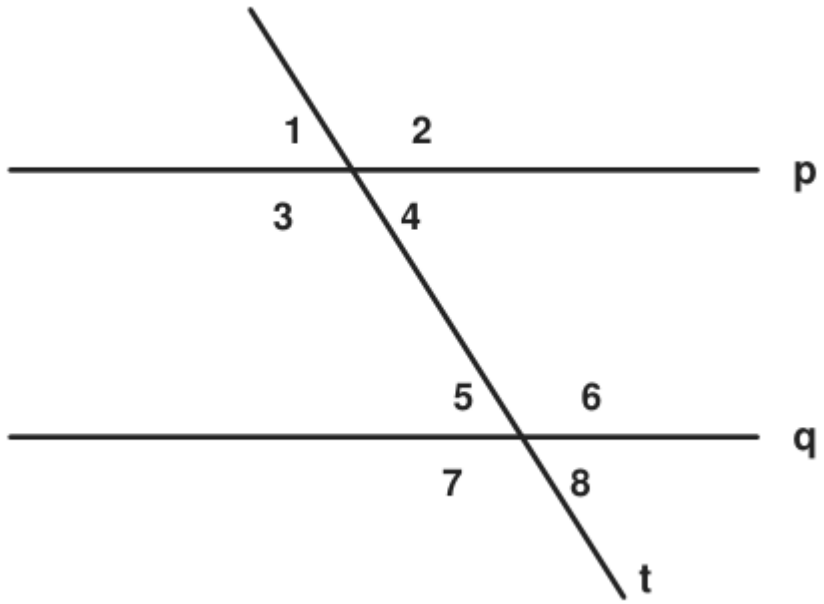
**REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-M-Q5**
Logic, Set 43: Proofs about lines and angles · MAIN · Q5/5 · Outcome LOGIC.PROOF.05 (local) · MP3

Two angles form a linear pair (together they make a straight line). One measures 48 degrees. What does the other measure, in degrees?

Degrees: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-B1-Q1**
Logic, Set 43: Proofs about lines and angles · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Angles 1 and 4 are vertical angles.

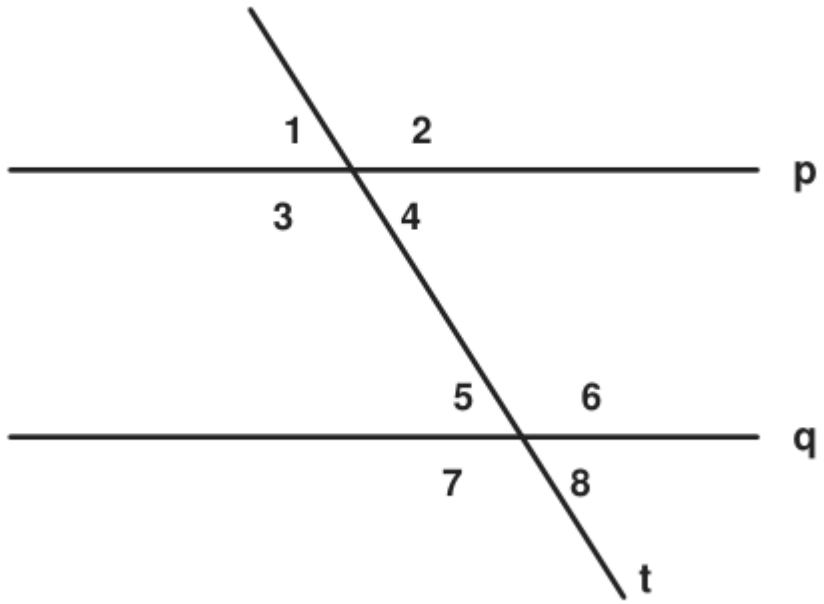



**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-B1-Q2**
Logic, Set 43: Proofs about lines and angles · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.PROOF.05 (local) · MP3

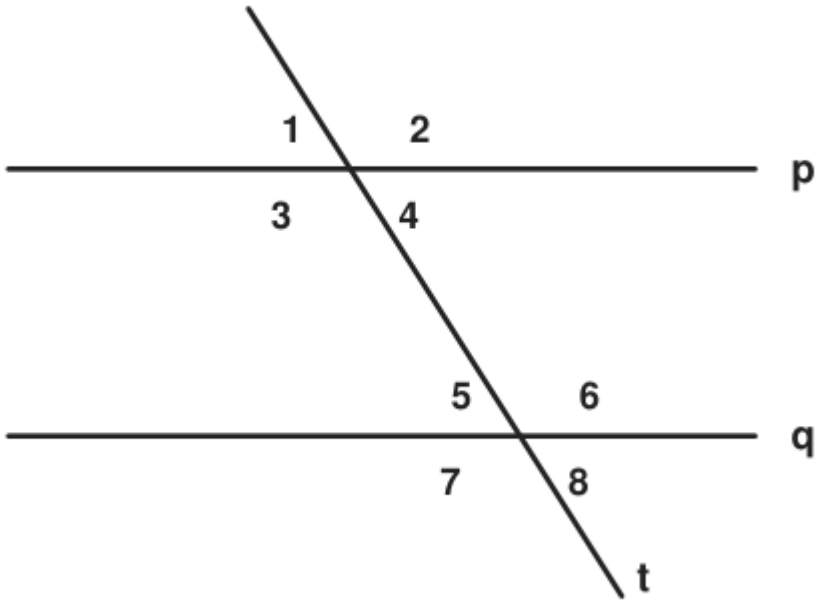
In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). What kind of angle pair are angles 3 and 7?




**A.** Vertical angles
**B.** A linear pair
**C.** Corresponding angles
**D.** Alternate interior angles

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-B1-Q3**
Logic, Set 43: Proofs about lines and angles · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Angles 3 and 5 are alternate interior angles.

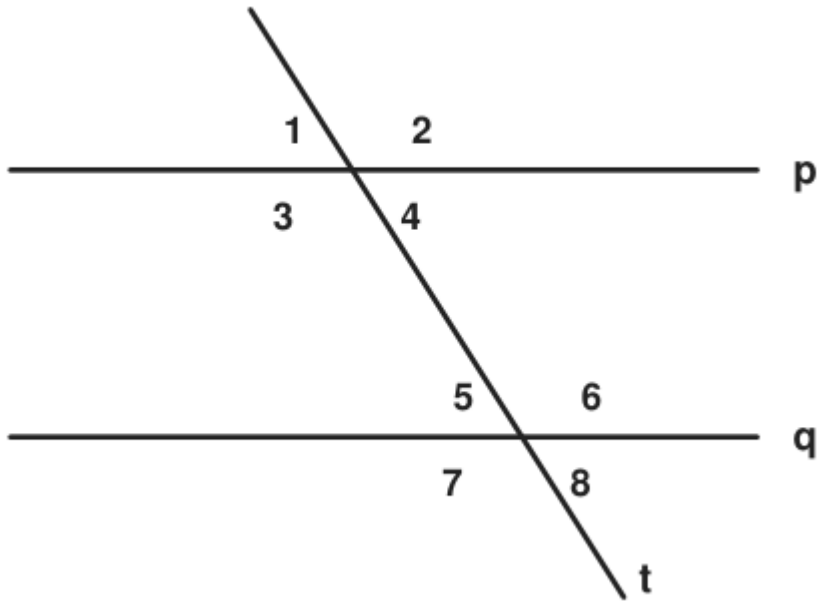



**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-B1-Q4**
Logic, Set 43: Proofs about lines and angles · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Angle 3 = 2x + 10 degrees and angle 2 = 3x - 15 degrees. They are vertical angles. Find x.




x = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-B1-Q5**
Logic, Set 43: Proofs about lines and angles · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.PROOF.05 (local) · MP3

M is the midpoint of segment AB, and line PM is perpendicular to AB, so P is on the perpendicular bisector of AB. Put the steps of the proof that PA = PB in a correct order. (Two orders are correct.)

Put these in order:

**A** AM = BM, and angles PMA and PMB are right angles (midpoint; perpendicular)
**B** PM = PM (a segment equals itself)
**C** Triangle PMA is congruent to triangle PMB (SAS)
**D** PA = PB (corresponding parts of congruent triangles)

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-F1-Q1**
Logic, Set 43: Proofs about lines and angles · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.PROOF.05 (local) · MP3

Theorem: a point is on the perpendicular bisector of AB exactly when it is the same distance from A and B. Point P has PA = PB = 7. What follows?

**A.** Nothing follows without more information.

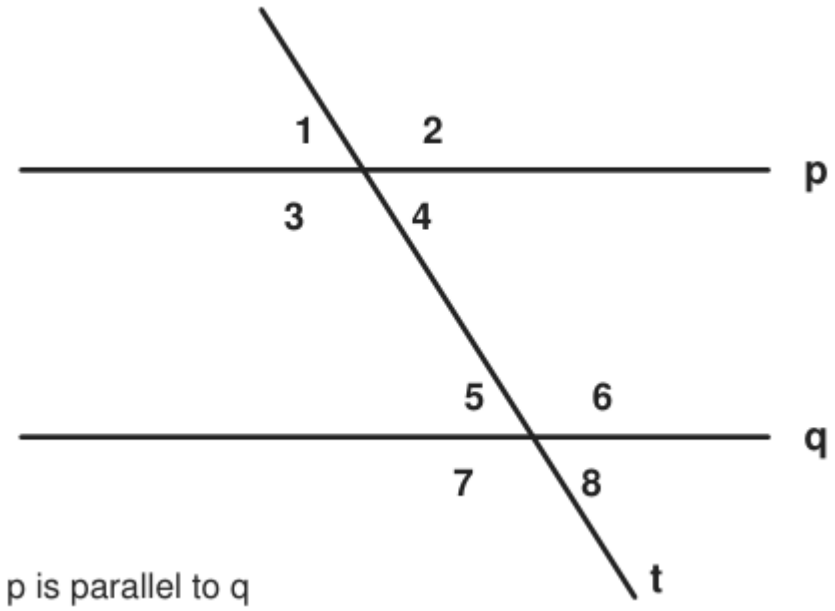
**B.** P is the midpoint of AB.

**C.** AB = 7.

**D.** P is on the perpendicular bisector of AB.

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S43-F1-Q2**
Logic, Set 43: Proofs about lines and angles · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Lines p and q are parallel. Then angle 4 + angle 6 = 180 degrees.

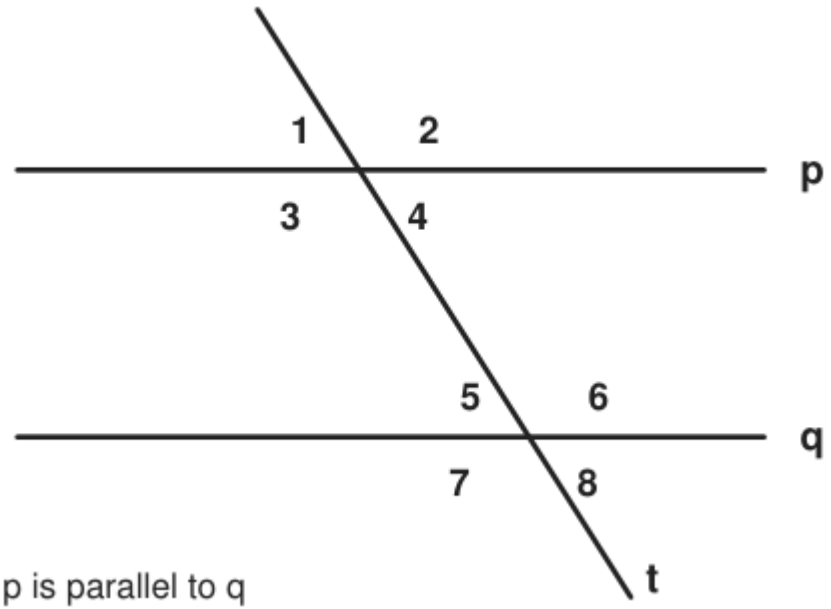



**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-F1-Q3**
Logic, Set 43: Proofs about lines and angles · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Lines p and q are parallel. Angle 4 = 3x + 5 degrees and angle 5 = 5x - 35 degrees. Find x.

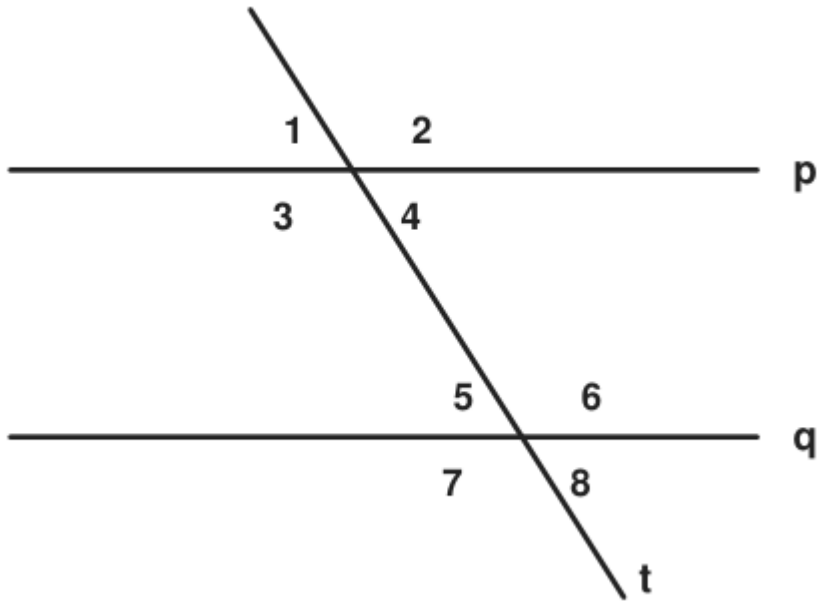



p is parallel to q

x = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-F1-Q4**
Logic, Set 43: Proofs about lines and angles · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Here p and q are NOT known to be parallel. Which fact would prove that they are parallel?




**A.** angle 3 = angle 6

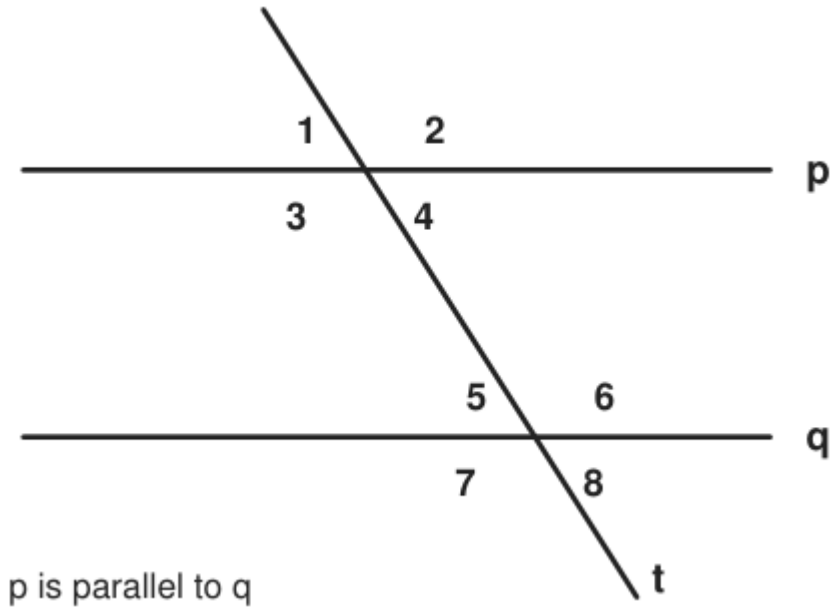
**B.** angle 1 + angle 2 = 180 degrees

**C.** angle 1 = angle 4

**D.** angle 5 = angle 8

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-F1-Q5**
Logic, Set 43: Proofs about lines and angles · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Put the steps of the proof that angle 4 + angle 6 = 180 degrees when p is parallel to q in a correct order. (Two orders are correct.)




p is parallel to q

Put these in order:

**A** angle 4 = angle 5 (alternate interior angles, p parallel to q)

**B** angle 5 + angle 6 = 180 degrees (linear pair)

**C** angle 4 + angle 6 = 180 degrees (substitute angle 4 for angle 5)

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-F2-Q1**
Logic, Set 43: Proofs about lines and angles · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.PROOF.05 (local) · MP3

A student writes: "Angle 1 = angle 4 because they look the same size in the diagram." What is wrong with this step?

**A.** A diagram is not a reason; the step needs the vertical angles theorem.

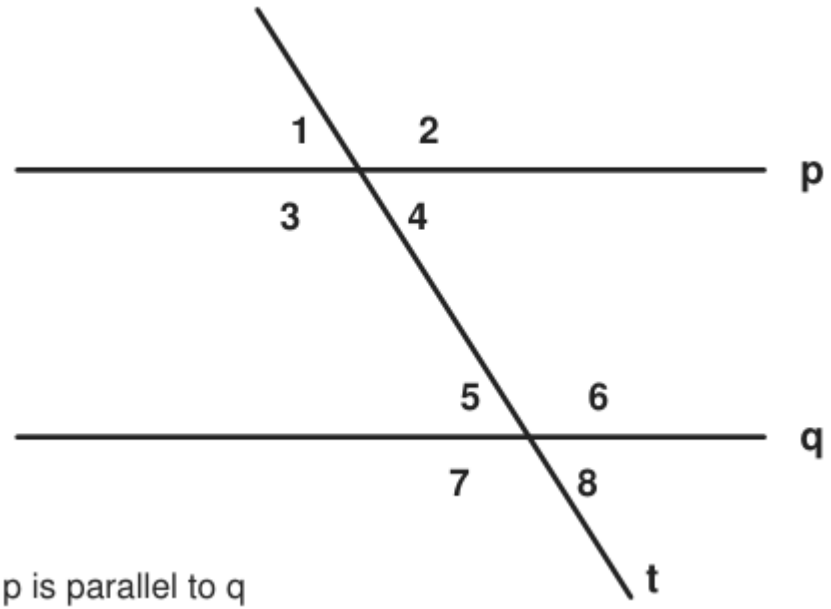
**B.** The step should say angle 1 + angle 4 = 180 degrees.

**C.** Nothing; looking at the diagram is enough.

**D.** Angles 1 and 4 are never equal.

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S43-F2-Q2**
Logic, Set 43: Proofs about lines and angles · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Lines p and q are parallel. Angle 1 = 2x + 30 degrees and angle 6 = x + 15 degrees. Find x.

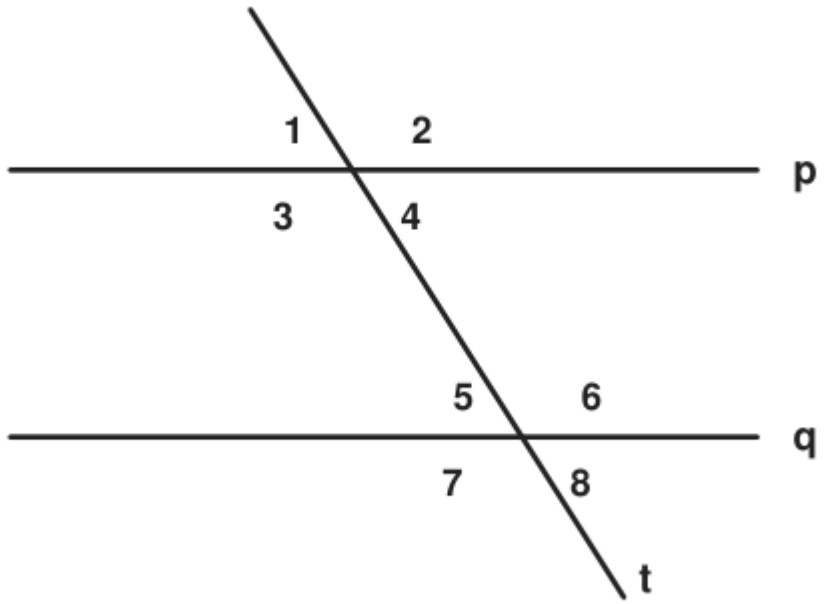



p is parallel to q

x = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-F2-Q3**
Logic, Set 43: Proofs about lines and angles · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.PROOF.05 (local) · MP3

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Without knowing that p and q are parallel, one can conclude that angle 3 = angle 6.




**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-F2-Q4**
Logic, Set 43: Proofs about lines and angles · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.PROOF.05 (local) · MP3

P is on the perpendicular bisector of segment AB. PA = 3x - 4 and PB = x + 10. What is PA?

PA = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S43-F2-Q5**
Logic, Set 43: Proofs about lines and angles · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.PROOF.05 (local) · MP3

# Set 44: Invariants and impossibility

Local outcome LOGIC.INVARIANT.02: Find a quantity that never changes under the allowed moves and use it to prove a target is impossible

Puzzle set · genre: invariants · MP7, MP3

MAIN: Find a quantity that never changes under the allowed moves and use it to prove a target is impossible

BACKWARD 1: Strategy: look for something that never changes

FORWARD 1: ADVANCED APPLICATION: Invariants with remainders

FORWARD 2: EXPERT CHALLENGE: Prove targets impossible

The numbers 1, 2, 3, ..., 10 are written on a board. A move erases any two numbers and writes their sum. After 9 moves one number is left. What is it?

**A.** It depends on the moves chosen

**B.** 55

**C.** 10

**D.** 45

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-M-Q1**
Logic, Set 44: Invariants and impossibility · MAIN · Q1/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

The numbers 1, 2, ..., 10 are written on a board. A move erases two numbers and writes their difference (larger minus smaller). After 9 moves one number is left. It can be 0.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-M-Q2**
Logic, Set 44: Invariants and impossibility · MAIN · Q2/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

Seven coins lie in a row: H H T H T T T (H = heads, T = tails). A move turns over any two coins that are next to each other. What is the smallest number of heads that can ever be showing?

Heads: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-M-Q3**
Logic, Set 44: Invariants and impossibility · MAIN · Q3/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

Seven coins lie in a row: H H T H T T T. A move turns over any two coins that are next to each other. Give a number of heads (0 to 7) that can be showing after some moves.

Heads = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S44-M-Q4**
Logic, Set 44: Invariants and impossibility · MAIN · Q4/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

A robot starts at the point (0, 0). Each move takes it from (x, y) to (x + 1, y + 2) or to (x + 2, y + 1). Which point can it reach?

**A.** (5, 5)
**B.** (2, 3)
**C.** (4, 5)
**D.** (7, 0)

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-M-Q5**
Logic, Set 44: Invariants and impossibility · MAIN · Q5/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

A counter shows 12. A move adds 4 or subtracts 6. Can it ever show 15?

**A.** No: 15 is not a multiple of 4.

**B.** Yes: after enough moves.

**C.** Yes: add 4, then subtract 1.

**D.** No: every move keeps the number even, and 15 is odd.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-B1-Q1**
Logic, Set 44: Invariants and impossibility · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

Three numbers on a board are 2, 4 and 9. A move adds 2 to any one of them. The total of the three numbers always stays odd.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-B1-Q2**
Logic, Set 44: Invariants and impossibility · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

A move erases two numbers a and b on a board and writes a + b. The board starts with 3, 5, 6 and 8. What number is left after 3 moves?

Number: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-B1-Q3**
Logic, Set 44: Invariants and impossibility · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

If a quantity is the same after every allowed move, it can never change, however many moves are made.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-B1-Q4**
Logic, Set 44: Invariants and impossibility · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

A robot starts at (0, 0). Each move goes from (x, y) to (x + 2, y) or to (x, y + 2). Which point can it NEVER reach?

**A.** (0, 6)
**B.** (3, 4)
**C.** (2, 4)
**D.** (4, 0)

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-B1-Q5**
Logic, Set 44: Invariants and impossibility · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

The numbers 1, 2, ..., 10 are on a board. A move erases two numbers a and b and writes a + b + 1. After 9 moves one number is left. What is it?

Number: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-F1-Q1**
Logic, Set 44: Invariants and impossibility · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

A robot starts at (0, 0). Each move goes from (x, y) to (x + 1, y + 2) or to (x + 2, y + 1). It can reach (10, 10).

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-F1-Q2**
Logic, Set 44: Invariants and impossibility · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

Three numbers are on a board: 1, 2, 3. A move picks two of them, adds 1 to one and subtracts 1 from the other. Which quantity never changes?

**A.** The product of the three numbers

**B.** The number of odd numbers

**C.** The total of the three numbers

**D.** The largest number

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S44-F1-Q3**
Logic, Set 44: Invariants and impossibility · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

Ten coins all show heads. A move turns over any four coins (they need not be next to each other). Give a number of heads from 0 to 10 that can NEVER be showing.

Heads = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-F1-Q4**
Logic, Set 44: Invariants and impossibility · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

A counter shows 100. A move subtracts 7 or adds 3. What is the smallest number of moves that makes it show 0?

Moves: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S44-F1-Q5**
Logic, Set 44: Invariants and impossibility · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

The numbers 1, 2, 3, 4 and 5 are on a board. A move erases two numbers a and b and writes a x b + a + b. After 4 moves one number is left. What is it?

Number: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-F2-Q1**
Logic, Set 44: Invariants and impossibility · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

A 4 by 4 grid of lights starts all off. A move switches every light in one whole row, or every light in one whole column. Some sequence of moves leaves exactly one light on.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-F2-Q2**
Logic, Set 44: Invariants and impossibility · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

A robot starts at (0, 0). Each move goes from (x, y) to (x + 3, y), to (x, y + 3), or to (x + 1, y + 1). Which point can it reach?

**A.** (2, 3)
**B.** (4, 0)
**C.** (1, 5)
**D.** (5, 2)

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-F2-Q3**
Logic, Set 44: Invariants and impossibility · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

A 4 by 4 grid of lights starts all off. A move switches every light in one whole row, or every light in one whole column. Give a number of lit lights from 1 to 15 that some sequence of moves can leave.

Lit lights = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S44-F2-Q4**
Logic, Set 44: Invariants and impossibility · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

The numbers 1, 2, 3, ..., 10 are on a board. A move erases two numbers a and b and writes a + b + 1. The last number left does not depend on which moves are made.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S44-F2-Q5**
Logic, Set 44: Invariants and impossibility · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.INVARIANT.02 (local) · MP7, MP3

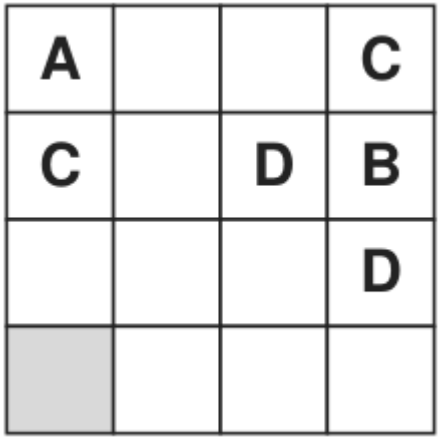
# Set 45: Letter grids and word squares

Local outcome LOGIC.GRID.02: Fill and analyze letter grids under stated rules and a supplied word list

Puzzle set · genre: letter grids · MP1, MP7

MAIN: Fill and analyze letter grids under stated rules and a supplied word list
BACKWARD 1: Strategy: fill the most constrained cell first
FORWARD 1: ADVANCED APPLICATION: Word squares from a list
FORWARD 2: EXPERT CHALLENGE: Prove a grid is impossible or count it

Rule: each row and each column must contain each of the letters A, B, C and D exactly once. The grid in the figure has six letters filled in. Which letter belongs in the shaded cell (bottom row, first column)?




Letter: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-M-Q1**
Logic, Set 45: Letter grids and word squares · MAIN · Q1/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Word list (the ONLY words allowed): CAT, ARE, TEN, ART, EAR, NET, TAN, ATE, RAT, ERA, TEA, ACE, ONE, NAP, APE, PEN, OAR, ORE, ROE, TOE, TON, NOT, ANT. In a word square, the three rows, read left to right, are the same words as the three columns, read top to bottom, in the same order. The first row of a word square is TEN. Give the second and third rows.

Row 2 = _____

Row 3 = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-M-Q2**
Logic, Set 45: Letter grids and word squares · MAIN · Q2/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

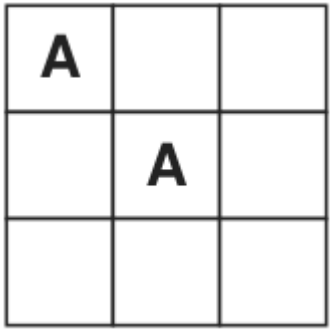
Rule: a 3 by 3 grid is filled with the letters A and B so that no two cells that share a side hold the same letter. The center cell holds A. Then all four corner cells hold the same letter.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S45-M-Q3**
Logic, Set 45: Letter grids and word squares · MAIN · Q3/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: each row and each column must contain each of the letters A, B and C exactly once. The top-left cell holds A and the center cell holds A. Which letter must go in the bottom-right cell?




**A.** A

**B.** B

**C.** It could be B or C

**D.** C

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-M-Q4**
Logic, Set 45: Letter grids and word squares · MAIN · Q4/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: each row and each column must contain each of the letters A, B and C exactly once. The first row is A B C. How many ways are there to complete the grid?

Ways: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S45-M-Q5**
Logic, Set 45: Letter grids and word squares · MAIN · Q5/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: each row and each column must contain each of the letters A, B and C exactly once. A row reads A _ C. Which letter fills the gap?

Letter: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S45-B1-Q1**
Logic, Set 45: Letter grids and word squares · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: each row and each column must contain each of the letters A, B and C exactly once. If column 1 holds A and B, its third cell must hold C.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S45-B1-Q2**
Logic, Set 45: Letter grids and word squares · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: each row and each column must contain each of the letters A, B, C and D exactly once. A cell's row already holds A and D, and its column holds B. Which letter must go in the cell?

**A.** A
**B.** D
**C.** B
**D.** C

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S45-B1-Q3**
Logic, Set 45: Letter grids and word squares · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: each row and each column must contain each of the letters A, B and C exactly once. A grid can have the row A B C directly above the row A C B.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-B1-Q4**
Logic, Set 45: Letter grids and word squares · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: each row and each column must contain each of the letters A, B and C exactly once. Rows 1 and 2 are A B C and C A B. What is row 3? Type the three letters with no spaces.

Row 3: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S45-B1-Q5**
Logic, Set 45: Letter grids and word squares · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Word list (the ONLY words allowed): CAT, ARE, TEN, ART, EAR, NET, TAN, ATE, RAT, ERA, TEA, ACE, ONE, NAP, APE, PEN, OAR, ORE, ROE, TOE, TON, NOT, ANT. In a word square, the three rows, read left to right, are the same words as the three columns, read top to bottom, in the same order. The first row of a word square is ATE. Give the second and third rows.

Row 2 = _____

Row 3 = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-F1-Q1**
Logic, Set 45: Letter grids and word squares · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Word list (the ONLY words allowed): CAT, ARE, TEN, ART, EAR, NET, TAN, ATE, RAT, ERA, TEA, ACE, ONE, NAP, APE, PEN, OAR, ORE, ROE, TOE, TON, NOT, ANT. In a word square, the three rows, read left to right, are the same words as the three columns, read top to bottom, in the same order. How many words on the list can be the first row of at least one word square?

Words: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-F1-Q2**
Logic, Set 45: Letter grids and word squares · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

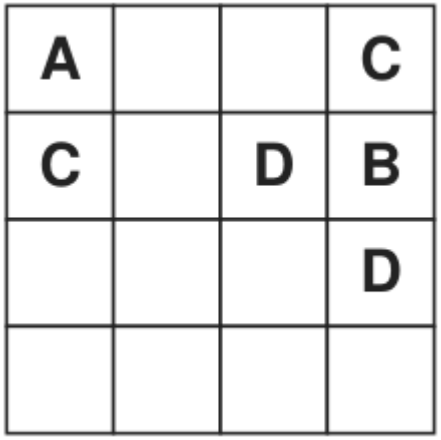
Word list (the ONLY words allowed): CAT, ARE, TEN, ART, EAR, NET, TAN, ATE, RAT, ERA, TEA, ACE, ONE, NAP, APE, PEN, OAR, ORE, ROE, TOE, TON, NOT, ANT. In a word square, the three rows, read left to right, are the same words as the three columns, read top to bottom, in the same order. A word square can start with the row ONE.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-F1-Q3**
Logic, Set 45: Letter grids and word squares · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: each row and each column must contain each of the letters A, B, C and D exactly once. The grid in the figure has six letters filled in. Type the complete top row (four letters, no spaces).




Row 1: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-F1-Q4**
Logic, Set 45: Letter grids and word squares · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: a 2 by 3 grid is filled with A and B so that no two cells sharing a side hold the same letter. How many such grids are there?

**A.** 2

**B.** 4

**C.** 6

**D.** 1

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S45-F1-Q5**
Logic, Set 45: Letter grids and word squares · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: each row and each column must contain each of the letters A, B, C and D exactly once. How many ways are there to fill the whole grid if the first row is A B C D?

Ways: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S45-F2-Q1**
Logic, Set 45: Letter grids and word squares · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: a 3 by 3 grid is filled with A and B so that no two cells sharing a side hold the same letter. The top-left cell holds A and the top-right cell holds B.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S45-F2-Q2**
Logic, Set 45: Letter grids and word squares · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: each row and each column must contain each of the letters A, B and C exactly once. Which partly filled grid can NOT be completed?

**A.** Row 1 starts A, and row 2 starts B

**B.** Row 1 starts A B, and row 2 ends in C

**C.** The diagonal from top left to bottom right is A, A, A

**D.** Row 1 is A B C

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S45-F2-Q3**
Logic, Set 45: Letter grids and word squares · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Word list (the ONLY words allowed): CAT, ARE, TEN, ART, EAR, NET, TAN, ATE, RAT, ERA, TEA, ACE, ONE, NAP, APE, PEN, OAR, ORE, ROE, TOE, TON, NOT, ANT. In a word square, the three rows, read left to right, are the same words as the three columns, read top to bottom, in the same order. Give the second and third rows of the word square whose first row is ROE.

Row 2 = _____

Row 3 = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S45-F2-Q4**
Logic, Set 45: Letter grids and word squares · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

Rule: each row and each column must contain each of the letters A, B and C exactly once. How many ways are there to fill an empty 3 by 3 grid?

Ways: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S45-F2-Q5**
Logic, Set 45: Letter grids and word squares · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.GRID.02 (local) · MP1, MP7

# Set 46: Explanations and missing assumptions

Local outcome LOGIC.EVIDENCE.01: Identify the unstated assumption an argument needs and evaluate competing explanations

Formal logic set · genre: formal logic · MP3

MAIN: Identify the unstated assumption an argument needs and evaluate competing explanations
BACKWARD 1: Find the gap between premise and conclusion
FORWARD 1: ADVANCED APPLICATION: Weaken and strengthen explanations
FORWARD 2: EXPERT CHALLENGE: Judge explanations with care

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "The school's robot passed every test in the lab, so it will work at the competition." Which assumption does this argument need?

**A.** The robot failed some tests last year.

**B.** The robot was built by students.

**C.** Conditions at the competition are like those in the lab.

**D.** Other teams also tested their robots.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-M-Q1**
Logic, Set 46: Explanations and missing assumptions · MAIN · Q1/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Mia's plant grew faster after she moved it to the window, so sunlight made it grow faster." Which assumption does this argument need?

**A.** Nothing else that affects growth changed when the plant was moved.

**B.** The window faces south.

**C.** Mia has other plants indoors.

**D.** Plants need water to grow.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-M-Q2**
Logic, Set 46: Explanations and missing assumptions · MAIN · Q2/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Every member of the chess club is in grade 10. Jo is in grade 10. So Jo is in the chess club." This argument would become valid if we add the assumption "Everyone in grade 10 is in the chess club."

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-M-Q3**
Logic, Set 46: Explanations and missing assumptions · MAIN · Q3/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

The cafeteria sold fewer sandwiches this week than last week. Which explanation is best supported if we also learn that the cafeteria was closed on two days this week?

**A.** Fewer students attend the school this year.

**B.** The cafeteria was open on fewer days to sell sandwiches.

**C.** Students stopped liking sandwiches.

**D.** The sandwich price went up.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-M-Q4**
Logic, Set 46: Explanations and missing assumptions · MAIN · Q4/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "This bridge is safe because it has stood for 80 years." The argument needs the assumption that the bridge will be used the same way as it has been.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-M-Q5**
Logic, Set 46: Explanations and missing assumptions · MAIN · Q5/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Sam is taller than Ana, so Sam is taller than Raj." Which assumption fills the gap?

**A.** Raj is taller than Ana.
**B.** Ana and Raj are friends.
**C.** Sam is the tallest in his class.
**D.** Ana is at least as tall as Raj.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-B1-Q1**
Logic, Set 46: Explanations and missing assumptions · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. Every true statement about the topic of an argument is an assumption that the argument needs.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-B1-Q2**
Logic, Set 46: Explanations and missing assumptions · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "The park was crowded on Saturday, so it will be crowded this Saturday." What is the gap?

**A.** From Saturday to Sunday
**B.** From the park to the city
**C.** From one past Saturday to a future Saturday
**D.** From crowded to empty

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-B1-Q3**
Logic, Set 46: Explanations and missing assumptions · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. If an assumption an argument needs turns out to be false, the conclusion no longer follows from the premises.

**A.** True
**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-B1-Q4**
Logic, Set 46: Explanations and missing assumptions · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "All squares are rectangles, so all rectangles are squares" needs no assumption to be valid.

**A.** True
**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-B1-Q5**
Logic, Set 46: Explanations and missing assumptions · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

Claim: "Students who eat breakfast score higher on tests, so eating breakfast raises test scores." Which fact would most WEAKEN the claim?

**A.** Test scores rose last year.
**B.** Many students eat breakfast.
**C.** Breakfast foods are cheap.
**D.** Students who eat breakfast also sleep more, and sleep is linked to higher scores.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-F1-Q1**
Logic, Set 46: Explanations and missing assumptions · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

Claim: "The new crosswalk signal reduced accidents at Elm Street." Which fact would most STRENGTHEN the claim?

**A.** The signal cost a lot of money.

**B.** Elm Street is near a school.

**C.** Accidents did not drop at similar streets that did not get the signal.

**D.** Drivers like the new signal.

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S46-F1-Q2**
Logic, Set 46: Explanations and missing assumptions · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "No reptiles have fur. This animal has fur. So this animal is not a reptile." This argument needs an unstated assumption to be valid.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-F1-Q3**
Logic, Set 46: Explanations and missing assumptions · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Our team lost every game we played in the rain. It will rain on Friday. So we will lose Friday's game." Which assumption does the argument need?

**A.** It rains more in the spring.

**B.** The rain caused the losses, rather than something else about those games.

**C.** Friday's game is at home.

**D.** Our team has played in the rain before.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-F1-Q4**
Logic, Set 46: Explanations and missing assumptions · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

Two explanations both fit the facts. Then no new fact can help decide between them.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-F1-Q5**
Logic, Set 46: Explanations and missing assumptions · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

A town's ice-cream sales and its number of sunburns rise and fall together during the year. Which explanation is best?

**A.** Hot, sunny weather increases both.
**B.** Eating ice cream causes sunburn.
**C.** Sunburns make people buy ice cream.
**D.** The two numbers match by chance every year.

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S46-F2-Q1**
Logic, Set 46: Explanations and missing assumptions · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Ming read the book, so Ming knows how it ends." Which assumption is needed?

**A.** Ming liked the book.
**B.** Ming read the book all the way to the end.
**C.** The book has a surprise ending.
**D.** The book is long.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-F2-Q2**
Logic, Set 46: Explanations and missing assumptions · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Every prize winner practiced daily. Kai practiced daily. So Kai won a prize." Adding "Only prize winners practiced daily" makes the argument valid.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-F2-Q3**
Logic, Set 46: Explanations and missing assumptions · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

A survey of the chess club finds that 90% of members prefer quiet study rooms. A student concludes that 90% of the whole school prefers quiet study rooms. What assumption does this need?

**A.** The survey asked about study rooms.

**B.** The chess club has more than 10 members.

**C.** Chess club members have the same preferences as the whole school.

**D.** Quiet rooms are better for studying.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-F2-Q4**
Logic, Set 46: Explanations and missing assumptions · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

If a needed assumption is shown to be false, the conclusion of the argument must be false.

**A.** True
**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S46-F2-Q5**
Logic, Set 46: Explanations and missing assumptions · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.EVIDENCE.01 (local) · MP3

# Set 47: Is there enough information?

Local outcome LOGIC.EVIDENCE.02: Decide whether the given information determines the answer, and what extra fact would

Formal logic set · genre: formal logic · MP1, MP3

MAIN: Decide whether the given information determines the answer, and what extra fact would

BACKWARD 1: Strategy: list what each clue leaves

FORWARD 1: ADVANCED APPLICATION: Which clue is enough?

FORWARD 2: EXPERT CHALLENGE: Design the missing clue

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) Ana, Ben and Cal ran a race with no ties. Who finished first? Clue (1): Ana finished ahead of Ben. Clue (2): Cal finished last.

**A.** Neither clue alone is enough, but both together are.
**B.** Clue (2) alone is enough, but clue (1) alone is not.
**C.** Clue (1) alone is enough, but clue (2) alone is not.
**D.** Each clue alone is enough.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-M-Q1**
Logic, Set 47: Is there enough information? · MAIN · Q1/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) x is a whole number from 1 to 20. What is x? Clue (1): x is a multiple of 7 and greater than 10. Clue (2): x is even.

**A.** Each clue alone is enough.

**B.** Neither clue alone is enough, but both together are.

**C.** Clue (2) alone is enough, but clue (1) alone is not.

**D.** Clue (1) alone is enough, but clue (2) alone is not.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-M-Q2**
Logic, Set 47: Is there enough information? · MAIN · Q2/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Two whole numbers, each from 1 to 9, add up to 17. Knowing only this is enough to know which two numbers they are.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S47-M-Q3**
Logic, Set 47: Is there enough information? · MAIN · Q3/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) A rectangle has whole-number side lengths. What is its area? Clue (1): its perimeter is 12. Clue (2): one side is twice as long as the other.

**A.** Clue (1) alone is enough, but clue (2) alone is not.

**B.** Neither clue alone is enough, but both together are.

**C.** Clue (2) alone is enough, but clue (1) alone is not.

**D.** Each clue alone is enough.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-M-Q4**
Logic, Set 47: Is there enough information? · MAIN · Q4/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

x is a whole number from 1 to 50. You know that x is a multiple of 6 and also a multiple of 4. How many values of x are still possible?

Values: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-M-Q5**
Logic, Set 47: Is there enough information? · MAIN · Q5/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

n is a whole number from 1 to 10, and n is a multiple of 4. How many values of n are possible?

Values: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-B1-Q1**
Logic, Set 47: Is there enough information? · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

n is a whole number from 1 to 10. Knowing that n is prime and even is enough to find n.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-B1-Q2**
Logic, Set 47: Is there enough information? · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Two whole numbers add to 10. Knowing only this is enough to find the larger number.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S47-B1-Q3**
Logic, Set 47: Is there enough information? · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

A clue is "enough" to answer a question when...

**A.** exactly one answer is left after using it.
**B.** at least one answer is left after using it.
**C.** it is true.
**D.** it rules out at least one answer.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-B1-Q4**
Logic, Set 47: Is there enough information? · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

n is a whole number from 1 to 20. Clue: n is a square number and n is odd. How many values of n are left?

Values: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S47-B1-Q5**
Logic, Set 47: Is there enough information? · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) Four friends sit in a row of four seats numbered 1 to 4: Dee, Eli, Fay and Gus. Who sits in seat 1? Clue (1): Dee sits in seat 4. Clue (2): Eli sits directly left of Fay, and Gus sits directly left of Eli.

**A.** Clue (2) alone is enough, but clue (1) alone is not.

**B.** Clue (1) alone is enough, but clue (2) alone is not.

**C.** Neither clue alone is enough, but both together are.

**D.** Each clue alone is enough.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-F1-Q1**
Logic, Set 47: Is there enough information? · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) Pat is older than Quinn. How old is Quinn, in whole years? Clue (1): their ages add to 30. Clue (2): Pat is 4 years older than Quinn.

**A.** Each clue alone is enough.

**B.** Clue (2) alone is enough, but clue (1) alone is not.

**C.** Clue (1) alone is enough, but clue (2) alone is not.

**D.** Neither clue alone is enough, but both together are.

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S47-F1-Q2**
Logic, Set 47: Is there enough information? · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) n is a whole number from 1 to 30. What is n? Clue (1): n leaves remainder 1 when divided by 7. Clue (2): n is a perfect square.

**A.** Each clue alone is enough.

**B.** Clue (1) alone is enough, but clue (2) alone is not.

**C.** Neither clue alone is enough, but both together are.

**D.** Clue (2) alone is enough, but clue (1) alone is not.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-F1-Q3**
Logic, Set 47: Is there enough information? · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

A bag holds only red and blue marbles, 12 in all. Knowing that there are 4 more red marbles than blue ones is enough to find the number of red marbles.

**A.** True
**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S47-F1-Q4**
Logic, Set 47: Is there enough information? · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

x is a whole number from 1 to 20 and x is a multiple of 3. Give one extra clue as a single number: a value m from 2 to 10 such that knowing "x is also a multiple of m" leaves exactly one possible x.

m = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S47-F1-Q5**
Logic, Set 47: Is there enough information? · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) n is a whole number from 1 to 20. What is n? Clue (1): n + 5 = 12. Clue (2): 2n = 14.

**A.** Clue (2) alone is enough, but clue (1) alone is not.

**B.** Clue (1) alone is enough, but clue (2) alone is not.

**C.** Neither clue alone is enough, but both together are.

**D.** Each clue alone is enough.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-F2-Q1**
Logic, Set 47: Is there enough information? · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) n is a whole number from 1 to 20. What is n? Clue (1): n is odd. Clue (2): n is a multiple of 13.

**A.** Clue (2) alone is enough, but clue (1) alone is not.

**B.** Clue (1) alone is enough, but clue (2) alone is not.

**C.** Neither clue alone is enough, but both together are.

**D.** Each clue alone is enough.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-F2-Q2**
Logic, Set 47: Is there enough information? · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Two whole numbers from 1 to 9 multiply to 12. Give a sum s such that also knowing "the two numbers add to s" tells you exactly which two numbers they are.

s = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S47-F2-Q3**
Logic, Set 47: Is there enough information? · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Two whole numbers, each from 1 to 9, multiply to 36. Knowing only this is enough to know which two numbers they are.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S47-F2-Q4**
Logic, Set 47: Is there enough information? · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

Three different whole numbers from 1 to 9 add up to 23. How many different sets of three numbers are possible?

Sets: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S47-F2-Q5**
Logic, Set 47: Is there enough information? · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.EVIDENCE.02 (local) · MP1, MP3

# Set 48: Digital displays

Local outcome LOGIC.SPATIAL.06: Reason about seven-segment digits: segments used, rotations and segment moves

Puzzle set · genre: seven-segment displays · MP7, MP6

MAIN: Reason about seven-segment digits: segments used, rotations and segment moves
BACKWARD 1: Read the segments of each digit
FORWARD 1: ADVANCED APPLICATION: Turned displays and segment counts
FORWARD 2: EXPERT CHALLENGE: Segment moves and best displays

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. How many segments are lit when the display shows 2026?



Segments: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S48-M-Q1**
Logic, Set 48: Digital displays · MAIN · Q1/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Which digit uses exactly 3 lit segments?




**A.** 4
**B.** 7
**C.** 1
**D.** 2

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-M-Q2**
Logic, Set 48: Digital displays · MAIN · Q2/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. When the display showing 69 is turned upside down, it shows 69 again.



**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-M-Q3**
Logic, Set 48: Digital displays · MAIN · Q3/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Give a digit that looks exactly the same after the display is turned upside down.



Digit = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S48-M-Q4**
Logic, Set 48: Digital displays · MAIN · Q4/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Exactly one lit segment of the digit 2 is moved to an unlit position. Which digit can result?



**A.** 5
**B.** 7
**C.** 8
**D.** 3

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S48-M-Q5**
Logic, Set 48: Digital displays · MAIN · Q5/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. How many segments does the digit 4 light?



Segments: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S48-B1-Q1**
Logic, Set 48: Digital displays · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. The digit 8 lights all seven segments.



**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-B1-Q2**
Logic, Set 48: Digital displays · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Which digit lights the fewest segments?



**A.** 4
**B.** 0
**C.** 1
**D.** 7

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-B1-Q3**
Logic, Set 48: Digital displays · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. The digits 0 and 6 light the same number of segments.



**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-B1-Q4**
Logic, Set 48: Digital displays · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. The digit 3 lights the middle segment and both left-hand segments.



**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-B1-Q5**
Logic, Set 48: Digital displays · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. The display shows 1958. How many segments are lit?



Segments: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S48-F1-Q1**
Logic, Set 48: Digital displays · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. The display shows 625. What does it show after being turned upside down?



**A.** 529

**B.** 526

**C.** 925

**D.** It does not show a number

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-F1-Q2**
Logic, Set 48: Digital displays · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. The display showing 1001 still shows a number after it is turned upside down.



**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-F1-Q3**
Logic, Set 48: Digital displays · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. How many two-digit numbers (10 to 99) still show a number after the display is turned upside down?



Numbers: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S48-F1-Q4**
Logic, Set 48: Digital displays · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Give a digit d such that adding exactly one lit segment to d (moving nothing) makes the digit 8.



d = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S48-F1-Q5**
Logic, Set 48: Digital displays · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Exactly one lit segment of the digit 3 is moved to an unlit position. Give a digit that can result.



Digit = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S48-F2-Q1**
Logic, Set 48: Digital displays · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. You may light at most 12 segments in total. What is the largest whole number you can show? (Any number of digits; the first digit is not 0.)



Number: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-F2-Q2**
Logic, Set 48: Digital displays · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Every digit can be turned into some other digit by moving exactly one lit segment.



**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-F2-Q3**
Logic, Set 48: Digital displays · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. A clock shows hours 1 to 12 and minutes 00 to 59 (for example 9:58 uses the digits 9, 5, 8). Which time lights the most segments?



**A.** 9:58

**B.** 10:08

**C.** 12:59

**D.** 11:11

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S48-F2-Q4**
Logic, Set 48: Digital displays · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. How many digits light more than 5 segments?



Digits: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S48-F2-Q5**
Logic, Set 48: Digital displays · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.SPATIAL.06 (local) · MP7, MP6

# Set 49: Common fallacies

Local outcome LOGIC.FALLACY.01: Recognize common informal fallacies in self-contained arguments

Formal logic set · genre: informal logic · MP3

MAIN: Recognize common informal fallacies in self-contained arguments
BACKWARD 1: Strategy: name what the argument does instead of giving reasons
FORWARD 1: ADVANCED APPLICATION: Straw men, slopes and popularity
FORWARD 2: EXPERT CHALLENGE: Is it really a fallacy?

Definitions: Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. Hasty generalization: drawing a broad conclusion from too few cases. Circular reasoning: using the conclusion, in other words, as a reason for itself. Appeal to popularity: claiming something is true because many people believe it. Dana argues that the library should stay open later. Raj replies: "Dana is always late to class, so her opinion about library hours does not count." Which fallacy is this?

**A.** Circular reasoning
**B.** Hasty generalization
**C.** Appeal to popularity
**D.** Attacking the person

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-M-Q1**
Logic, Set 49: Common fallacies · MAIN · Q1/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: False choice: presenting only two options when others are possible. Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. Straw man: attacking a distorted version of someone's argument instead of the real one. After, so because: concluding that A caused B only because B happened after A. "Either we cancel the field trip, or every student will get lost in the city." Which fallacy is this?

**A.** False choice
**B.** After, so because
**C.** Straw man
**D.** Slippery slope

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-M-Q2**
Logic, Set 49: Common fallacies · MAIN · Q2/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Hasty generalization: drawing a broad conclusion from too few cases. Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. False choice: presenting only two options when others are possible. Circular reasoning: using the conclusion, in other words, as a reason for itself. "I met two rude people from that town, so everyone from that town is rude." Which fallacy is this?

**A.** Attacking the person
**B.** Circular reasoning
**C.** Hasty generalization
**D.** False choice

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-M-Q3**
Logic, Set 49: Common fallacies · MAIN · Q3/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Circular reasoning: using the conclusion, in other words, as a reason for itself. "This app is the best because it is better than all the others" is an example of circular reasoning.

**A.** True
**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-M-Q4**
Logic, Set 49: Common fallacies · MAIN · Q4/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: After, so because: concluding that A caused B only because B happened after A. Hasty generalization: drawing a broad conclusion from too few cases. Appeal to popularity: claiming something is true because many people believe it. Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. "After I started wearing my lucky socks, our team won three games in a row. The socks made us win." Which fallacy is this?

**A.** Hasty generalization
**B.** After, so because
**C.** Slippery slope
**D.** Appeal to popularity

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-M-Q5**
Logic, Set 49: Common fallacies · MAIN · Q5/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Appeal to popularity: claiming something is true because many people believe it. Straw man: attacking a distorted version of someone's argument instead of the real one. False choice: presenting only two options when others are possible. Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. Which definition describes "Millions of people watch this show, so it must be educational"?

**A.** Straw man
**B.** Appeal to popularity
**C.** Slippery slope
**D.** False choice

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-B1-Q1**
Logic, Set 49: Common fallacies · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Hasty generalization: drawing a broad conclusion from too few cases. Concluding that all dogs are friendly after meeting three friendly dogs is a hasty generalization.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-B1-Q2**
Logic, Set 49: Common fallacies · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. Saying "The bridge design is unsafe because the beams are too thin for the load" is attacking the person.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-B1-Q3**
Logic, Set 49: Common fallacies · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

An argument's reason is about who made a claim rather than about the claim. Which question best helps you notice this?

**A.** Is the reason about the claim, or about something else?
**B.** Do I agree with the conclusion?
**C.** Is the conclusion surprising?
**D.** Is the argument long?

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-B1-Q4**
Logic, Set 49: Common fallacies · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. "If we allow phones at lunch, then students will use them in class, then stop learning, then fail school" is a slippery slope.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-B1-Q5**
Logic, Set 49: Common fallacies · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Straw man: attacking a distorted version of someone's argument instead of the real one. Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. False choice: presenting only two options when others are possible. Hasty generalization: drawing a broad conclusion from too few cases. Ava says the school should add one vegetarian meal option. Ben replies: "Ava wants to ban meat from the cafeteria completely, and that is unfair to everyone." Which fallacy is this?

**A.** Hasty generalization
**B.** Attacking the person
**C.** Straw man
**D.** False choice

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-F1-Q1**
Logic, Set 49: Common fallacies · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. False choice: presenting only two options when others are possible. After, so because: concluding that A caused B only because B happened after A. Circular reasoning: using the conclusion, in other words, as a reason for itself. "If we let one student retake the quiz, soon everyone will retake every quiz, and grades will mean nothing." Which fallacy is this?

**A.** Circular reasoning
**B.** After, so because
**C.** False choice
**D.** Slippery slope

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-F1-Q2**
Logic, Set 49: Common fallacies · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Appeal to popularity: claiming something is true because many people believe it. "Most dentists in a large survey recommend brushing twice a day, so brushing twice a day is likely a good idea" is an appeal to popularity.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-F1-Q3**
Logic, Set 49: Common fallacies · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Circular reasoning: using the conclusion, in other words, as a reason for itself. Appeal to popularity: claiming something is true because many people believe it. Straw man: attacking a distorted version of someone's argument instead of the real one. Hasty generalization: drawing a broad conclusion from too few cases. "The rule must be fair because it is the rule, and rules are fair." Which fallacy is this?

**A.** Hasty generalization
**B.** Appeal to popularity
**C.** Circular reasoning
**D.** Straw man

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-F1-Q4**
Logic, Set 49: Common fallacies · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: After, so because: concluding that A caused B only because B happened after A. Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. False choice: presenting only two options when others are possible. "The new principal started in September, and in October the cafeteria food got worse. The principal ruined the food." Which fallacy is this?

**A.** Attacking the person
**B.** After, so because
**C.** False choice
**D.** Slippery slope

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-F1-Q5**
Logic, Set 49: Common fallacies · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. Which reply is NOT attacking the person?

**A.** "Your numbers are from 2010, and the prices have changed since then."

**B.** "Nobody listens to you anyway."

**C.** "You are too young to understand prices."

**D.** "You only say that because you work for the store."

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S49-F2-Q1**
Logic, Set 49: Common fallacies · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: False choice: presenting only two options when others are possible. "A whole number is either even or odd" is a false choice.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-F2-Q2**
Logic, Set 49: Common fallacies · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Hasty generalization: drawing a broad conclusion from too few cases. After, so because: concluding that A caused B only because B happened after A. Straw man: attacking a distorted version of someone's argument instead of the real one. Appeal to popularity: claiming something is true because many people believe it. Which argument avoids all four fallacies defined above?

**A.** My two cousins liked the plan, so most voters like it.

**B.** Everyone online says the plan is good, so it is good.

**C.** Our team won after the coach changed shoes, so the shoes helped.

**D.** A random sample of 500 voters favored the plan, so most voters probably favor it.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-F2-Q3**
Logic, Set 49: Common fallacies · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Straw man: attacking a distorted version of someone's argument instead of the real one. Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. Appeal to popularity: claiming something is true because many people believe it. Circular reasoning: using the conclusion, in other words, as a reason for itself. Mo says homework should be shorter on weekends. Lee replies: "So you think students should never study at all? That would be terrible." Which fallacy is this?

**A.** Slippery slope
**B.** Straw man
**C.** Appeal to popularity
**D.** Circular reasoning

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-F2-Q4**
Logic, Set 49: Common fallacies · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

Definitions: Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. An argument that predicts a chain of results is always a slippery slope, even when each step is supported by evidence.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S49-F2-Q5**
Logic, Set 49: Common fallacies · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.FALLACY.01 (local) · MP3

# Set 50: Weighing evidence

Local outcome LOGIC.EVIDENCE.03: Judge whether evidence supports a claim: samples, comparisons and correlation

Formal logic set · genre: informal logic · MP3, MP2

MAIN: Judge whether evidence supports a claim: samples, comparisons and correlation
BACKWARD 1: Compare rates, not counts
FORWARD 1: ADVANCED APPLICATION: Samples and comparisons
FORWARD 2: EXPERT CHALLENGE: Comparisons that reverse

To learn how often people in a town exercise, a student surveys 100 people leaving a gym. What is the main problem with this evidence?

**A.** People leaving a gym probably exercise more than the town as a whole.

**B.** The survey should have asked about diet too.

**C.** 100 people is always too few for a survey.

**D.** Surveys can never give useful evidence.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-M-Q1**
Logic, Set 50: Weighing evidence · MAIN · Q1/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

School A: 45 of its 60 students passed a test. School B: 52 of its 80 students passed. What percent of School A's students passed?

Percent: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-M-Q2**
Logic, Set 50: Weighing evidence · MAIN · Q2/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

In many towns, more ice-cream sales come with more swimming accidents. This shows that eating ice cream causes swimming accidents.

**A.** True
**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S50-M-Q3**
Logic, Set 50: Weighing evidence · MAIN · Q3/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

Which study design gives the best evidence that a tutoring program raises test scores?

**A.** Measure the scores of tutored students only.
**B.** Compare tutored students with students who chose not to sign up.
**C.** Randomly assign students to get tutoring or not, then compare the two groups' scores.
**D.** Ask tutored students whether they feel they improved.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-M-Q4**
Logic, Set 50: Weighing evidence · MAIN · Q4/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

A claim says that more than half of a school's students prefer a later start time. In a random sample of 40 students, n said they prefer it. Give a value of n that would support the claim.

n = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S50-M-Q5**
Logic, Set 50: Weighing evidence · MAIN · Q5/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

18 of 24 students in a class walk to school. What percent is that?

Percent: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-B1-Q1**
Logic, Set 50: Weighing evidence · BACKWARD 1 (prerequisite) · Q1/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

Club X has 30 members and 12 play chess. Club Y has 10 members and 6 play chess. A larger share of Club Y plays chess.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-B1-Q2**
Logic, Set 50: Weighing evidence · BACKWARD 1 (prerequisite) · Q2/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

Town P had 50 car accidents and Town Q had 30. What else do you need before saying Town P's roads are more dangerous?

**A.** The colors of the cars
**B.** Nothing; 50 is more than 30
**C.** The names of the drivers
**D.** How many people or cars each town has

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-B1-Q3**
Logic, Set 50: Weighing evidence · BACKWARD 1 (prerequisite) · Q3/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

A survey of 200 people finds that 30 own a bike. What percent own a bike?

Percent: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-B1-Q4**
Logic, Set 50: Weighing evidence · BACKWARD 1 (prerequisite) · Q4/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

If Group A has more members with a trait than Group B, then a larger percent of Group A has the trait.

**A.** True
**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-B1-Q5**
Logic, Set 50: Weighing evidence · BACKWARD 1 (prerequisite) · Q5/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

A website asks visitors to vote on whether the town needs a new park. Of 300 votes, 80% say yes. Why is this weak evidence about the whole town?

**A.** 80% is too high to be believed.

**B.** 300 votes is more than enough.

**C.** Websites cannot count votes.

**D.** People who choose to vote online may care more about parks than others.

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S50-F1-Q1**
Logic, Set 50: Weighing evidence · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

Class A: 21 of 28 students finished the project on time. Class B: 24 of 30. What is the higher of the two on-time percents?

Percent: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S50-F1-Q2**
Logic, Set 50: Weighing evidence · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

A study finds that students who own more books tend to have higher reading scores. This alone proves that buying books raises reading scores.

**A.** True
**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-F1-Q3**
Logic, Set 50: Weighing evidence · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

To test whether a new fertilizer helps tomato plants grow, which design is best?

**A.** Give fertilizer to a randomly chosen half of the plants and compare them with the other half.

**B.** Give fertilizer to all plants and measure them.

**C.** Ask a gardener whether the plants look bigger.

**D.** Use the fertilizer only on the healthiest plants.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-F1-Q4**
Logic, Set 50: Weighing evidence · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

A coin is flipped 50 times to test whether it is fair. Give a number of heads h from 0 to 50 that would be strong evidence the coin favors heads. Use this test: h is at least 35.

h = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S50-F1-Q5**
Logic, Set 50: Weighing evidence · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

Two clinics treat a condition. Clinic A treats 100 mild cases (90 recover) and 20 severe cases (8 recover). Clinic B treats 20 mild cases (19 recover) and 100 severe cases (50 recover). What percent of all Clinic A patients recover, to the nearest whole percent?

Percent: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S50-F2-Q1**
Logic, Set 50: Weighing evidence · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

In the clinics above, Clinic A has the higher overall recovery rate (about 82% to 58%). Which statement is true?

**A.** Clinic A is better for both mild and severe cases.

**B.** Clinic B has the higher recovery rate for mild cases AND for severe cases.

**C.** Clinic A must be the better clinic.

**D.** The two clinics are equally good for mild cases.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-F2-Q2**
Logic, Set 50: Weighing evidence · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

In the clinics above, comparing only the overall recovery rates gives a fair comparison of the clinics.

**A.** True
**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-F2-Q3**
Logic, Set 50: Weighing evidence · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

A test for a rare condition is positive for 95 of every 100 people who have it and for 5 of every 100 people who do not. In a town of 10,000 people, 100 have the condition. About how many people in the town test positive in all?

People: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S50-F2-Q4**
Logic, Set 50: Weighing evidence · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

In the town above, 590 people test positive and 95 of them have the condition. What fraction of the positive tests are correct?

**A.** About 5%
**B.** About 50%
**C.** About 16%
**D.** About 95%

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S50-F2-Q5**
Logic, Set 50: Weighing evidence · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.EVIDENCE.03 (local) · MP3, MP2

# Set 51: Best possible: constrained optimization

Local outcome LOGIC.OPTIMIZE.01: Find the best value under stated constraints and show nothing better is possible

Puzzle set · genre: optimization · MP1, MP7

MAIN: Find the best value under stated constraints and show nothing better is possible
BACKWARD 1: Strategy: find a good answer, then prove nothing beats it
FORWARD 1: ADVANCED APPLICATION: When the obvious choice is not best
FORWARD 2: EXPERT CHALLENGE: Prove the best value

Two whole numbers add up to 20. What is the largest possible product?

Product: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S51-M-Q1**
Logic, Set 51: Best possible: constrained optimization · MAIN · Q1/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

A rectangular pen uses exactly 24 meters of fence, and its sides are whole numbers of meters. What is the largest possible area, in square meters?

Area: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S51-M-Q2**
Logic, Set 51: Best possible: constrained optimization · MAIN · Q2/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

A 12-hour digital clock shows times from 1:00 to 12:59. Which time has the largest product of its digits?

**A.** 8:59
**B.** 9:59
**C.** 11:59
**D.** 12:59

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S51-M-Q3**
Logic, Set 51: Best possible: constrained optimization · MAIN · Q3/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Using coins worth 1, 5, 10 and 25 cents, the fewest coins that make exactly 30 cents is 2.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-M-Q4**
Logic, Set 51: Best possible: constrained optimization · MAIN · Q4/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Whole numbers x and y are each 0 or more, with x + 2y at most 10 and x at most 6. Make x + 3y as large as possible. Give x and y.

x = _____

y = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S51-M-Q5**
Logic, Set 51: Best possible: constrained optimization · MAIN · Q5/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Two whole numbers add to 14. What is the largest possible product?

Product: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S51-B1-Q1**
Logic, Set 51: Best possible: constrained optimization · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

To show that a value is the largest possible, it is enough to find one example that reaches it.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S51-B1-Q2**
Logic, Set 51: Best possible: constrained optimization · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Which numbers from the list 4, 9, 2, 7, 5 give the largest sum if you may choose any two?

**A.** 4 and 9
**B.** 9 and 5
**C.** 9 and 7
**D.** 7 and 5

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S51-B1-Q3**
Logic, Set 51: Best possible: constrained optimization · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Using coins worth 1, 5 and 10 cents, what is the fewest coins that make 27 cents?

Coins: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S51-B1-Q4**
Logic, Set 51: Best possible: constrained optimization · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

A rectangle with whole-number sides and perimeter 20 can have area 25.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S51-B1-Q5**
Logic, Set 51: Best possible: constrained optimization · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Coins are worth 1, 4 and 6 cents. What is the fewest coins that make exactly 8 cents?

Coins: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-F1-Q1**
Logic, Set 51: Best possible: constrained optimization · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Whole numbers add up to 10 (you choose how many numbers). What is the largest possible product?

Product: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S51-F1-Q2**
Logic, Set 51: Best possible: constrained optimization · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Coins are worth 1, 4 and 6 cents. Always taking the largest coin that fits gives the fewest coins for every amount.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-F1-Q3**
Logic, Set 51: Best possible: constrained optimization · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

A backpack can hold at most 10 kilograms. Items: tent 6 kg, value 8; stove 4 kg, value 5; food 5 kg, value 6; lamp 1 kg, value 3. Which choice has the greatest total value?

**A.** Food, stove, lamp

**B.** Food, lamp

**C.** Tent, stove

**D.** Tent, lamp

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S51-F1-Q4**
Logic, Set 51: Best possible: constrained optimization · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Two whole numbers are each at least 1 and their product is 36. Make their sum as small as possible. Give the two numbers.

a = _____

b = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S51-F1-Q5**
Logic, Set 51: Best possible: constrained optimization · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Whole numbers add up to 20 (you choose how many numbers). What is the largest possible product?

Product: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S51-F2-Q1**
Logic, Set 51: Best possible: constrained optimization · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Coins are worth 1, 7 and 10 cents. What is the fewest coins that make exactly 14 cents?

Coins: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S51-F2-Q2**
Logic, Set 51: Best possible: constrained optimization · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Among rectangles with whole-number sides and area 36, the 6 by 6 square has the smallest perimeter.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S51-F2-Q3**
Logic, Set 51: Best possible: constrained optimization · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Whole numbers a, b and c are each at least 1 and add up to 12. Make the product a x b x c as large as possible. Give a, b and c.

a = _____

b = _____

c = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-F2-Q4**
Logic, Set 51: Best possible: constrained optimization · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

Use each of the digits 1, 2, 3 and 4 once to make two 2-digit numbers. Which pair has the largest product?

**A.** 31 x 24
**B.** 43 x 21
**C.** 42 x 31
**D.** 41 x 32

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S51-F2-Q5**
Logic, Set 51: Best possible: constrained optimization · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.OPTIMIZE.01 (local) · MP1, MP7

# Set 52: Patterns from a stated rule family

Local outcome LOGIC.PATTERN.01: Identify the rule in a stated family that fits a sequence, and continue it

Puzzle set · genre: sequences · MP8, MP7

MAIN: Identify the rule in a stated family that fits a sequence, and continue it
BACKWARD 1: Strategy: write the differences
FORWARD 1: ADVANCED APPLICATION: Rules for letters and far terms
FORWARD 2: EXPERT CHALLENGE: Work with the rule as a formula

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). What is the next term of 3, 7, 11, 15, ...?

Next term: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S52-M-Q1**
Logic, Set 52: Patterns from a stated rule family · MAIN · Q1/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Which rule fits 2, 6, 18, 54, ...?

**A.** Multiply by 3 each time

**B.** Add 4, then 12, then 36

**C.** Add 4 each time

**D.** Multiply by 2 each time

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S52-M-Q2**
Logic, Set 52: Patterns from a stated rule family · MAIN · Q2/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). What is the next term of 1, 2, 4, 7, 11, ...?

Next term: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S52-M-Q3**
Logic, Set 52: Patterns from a stated rule family · MAIN · Q3/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). The sequence 5, 10, 20, 35, ... follows rule (3).

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-M-Q4**
Logic, Set 52: Patterns from a stated rule family · MAIN · Q4/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Letters are numbered by their place in the alphabet: A = 1, B = 2, ..., Z = 26. The letters C, F, I, L follow rule (1) by their numbers. Give the next letter.

Next letter = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S52-M-Q5**
Logic, Set 52: Patterns from a stated rule family · MAIN · Q5/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Write the differences of 2, 5, 9, 14. What is the last difference?

Difference: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S52-B1-Q1**
Logic, Set 52: Patterns from a stated rule family · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). The sequence 10, 7, 4, 1, ... follows rule (1).

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-B1-Q2**
Logic, Set 52: Patterns from a stated rule family · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Which rule fits 1, 4, 16, 64, ...?

**A.** None of the rules
**B.** Rule (2)
**C.** Rule (3)
**D.** Rule (1)

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S52-B1-Q3**
Logic, Set 52: Patterns from a stated rule family · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). What is the next term of 50, 45, 40, 35, ...?

Next term: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S52-B1-Q4**
Logic, Set 52: Patterns from a stated rule family · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). The sequence 3, 6, 12, 24, ... follows rule (1).

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-B1-Q5**
Logic, Set 52: Patterns from a stated rule family · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Letters are numbered by their place in the alphabet: A = 1, B = 2, ..., Z = 26. The letters B, D, G, K follow rule (3) by their numbers. Give the next letter.

Next letter = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S52-F1-Q1**
Logic, Set 52: Patterns from a stated rule family · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). What is the 10th term of 4, 7, 10, 13, ...?

10th term: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S52-F1-Q2**
Logic, Set 52: Patterns from a stated rule family · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Knowing only the first two terms, 3 and 6, is enough to know the third term.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-F1-Q3**
Logic, Set 52: Patterns from a stated rule family · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). The first three terms are 1, 2, 4. Which statement is true?

**A.** Only rule (3) fits, so the next term is 7.

**B.** Rule (1) fits, so the next term is 6.

**C.** Rule (2) and rule (3) both fit, giving 8 or 7 next.

**D.** Only rule (2) fits, so the next term is 8.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-F1-Q4**
Logic, Set 52: Patterns from a stated rule family · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). What is the 8th term of 2, 6, 18, 54, ...?

8th term: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S52-F1-Q5**
Logic, Set 52: Patterns from a stated rule family · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). The sequence is 5, 9, 13, 17, .... Give the 25th term.

25th term = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S52-F2-Q1**
Logic, Set 52: Patterns from a stated rule family · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). In the sequence 3, 6, 12, 24, ..., which term is the first one greater than 1000? Give its position (1 for the first term).

Position: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S52-F2-Q2**
Logic, Set 52: Patterns from a stated rule family · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). What is the 10th term of 1, 3, 6, 10, ...?

10th term: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S52-F2-Q3**
Logic, Set 52: Patterns from a stated rule family · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). A sequence that follows rule (1) with a positive amount added will eventually pass every number.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-F2-Q4**
Logic, Set 52: Patterns from a stated rule family · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Letters are numbered by their place in the alphabet: A = 1, B = 2, ..., Z = 26. Which letter sequence follows rule (2) by the letters' numbers?

**A.** A, B, D, G
**B.** A, C, E, G
**C.** B, C, E, H
**D.** A, B, D, H

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S52-F2-Q5**
Logic, Set 52: Patterns from a stated rule family · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.PATTERN.01 (local) · MP8, MP7

# Set 53: Rearranging letters

Local outcome LOGIC.WORD.03: Find anagrams and letter-bank words within a supplied word list and letter rules

Puzzle set · genre: anagrams · MP7, MP1

MAIN: Find anagrams and letter-bank words within a supplied word list and letter rules
BACKWARD 1: Strategy: sort the letters
FORWARD 1: ADVANCED APPLICATION: Letter banks and joined parts
FORWARD 2: EXPERT CHALLENGE: Phrases and hidden words

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same letters, each as many times, in a different order. Which word on the list is an anagram of LEMON?

**A.** SOLEMN
**B.** LANE
**C.** TONE
**D.** MELON

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-M-Q1**
Logic, Set 53: Rearranging letters · MAIN · Q1/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same letters, each as many times, in a different order. Give a word on the list that is an anagram of STARE.

Word = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-M-Q2**
Logic, Set 53: Rearranging letters · MAIN · Q2/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same letters, each as many times, in a different order. LISTEN, SILENT, ENLIST and TINSEL are all anagrams of one another.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-M-Q3**
Logic, Set 53: Rearranging letters · MAIN · Q3/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. How many words on the list can be spelled using the letters of PLANET, each letter at most once?

Words: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-M-Q4**
Logic, Set 53: Rearranging letters · MAIN · Q4/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. Read the letters STOPAN from right to left. Which word on the list appears as a block of neighboring letters in that backward reading?

**A.** SPOT
**B.** POTS
**C.** PAN
**D.** STOP

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-M-Q5**
Logic, Set 53: Rearranging letters · MAIN · Q5/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Write the letters of NOTE in alphabetical order (no spaces).

Letters: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S53-B1-Q1**
Logic, Set 53: Rearranging letters · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

An anagram uses exactly the same letters, each as many times, in a different order. STONE and NOTES are anagrams.

**A.** True
**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-B1-Q2**
Logic, Set 53: Rearranging letters · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

An anagram uses exactly the same letters, each as many times, in a different order. TASTE and STATE are anagrams.

**A.** True
**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-B1-Q3**
Logic, Set 53: Rearranging letters · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

An anagram uses exactly the same letters, each as many times, in a different order. Which word is NOT an anagram of SPOT?

**A.** POTS
**B.** STOP
**C.** SPOTS
**D.** TOPS

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S53-B1-Q4**
Logic, Set 53: Rearranging letters · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

An anagram uses exactly the same letters, each as many times, in a different order. BREAD and BEARD are anagrams.

**A.** True
**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-B1-Q5**
Logic, Set 53: Rearranging letters · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. Join one part from the first group to one part from the second group to make a word on the list. First group: PL, ST, HE. Second group: ART, ANET, ONE. Give the word.

Word = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-F1-Q1**
Logic, Set 53: Rearranging letters · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same letters, each as many times, in a different order. How many words on the list are anagrams of EARTH (not counting EARTH itself)?

Words: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-F1-Q2**
Logic, Set 53: Rearranging letters · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. You may use the letters S, T, O, P as often as you like. Which is the longest word on the list you can spell?

**A.** All of SPOT, STOP, POTS, TOPS and POST tie at 4 letters

**B.** SPOTS

**C.** STONE

**D.** STOP is the only longest word

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-F1-Q3**
Logic, Set 53: Rearranging letters · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same letters, each as many times, in a different order. NIGHT and THING are anagrams.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-F1-Q4**
Logic, Set 53: Rearranging letters · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same letters, each as many times, in a different order. Group the list words into anagram families (words with the same sorted letters). How many families have exactly 4 or more words?

Families: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-F1-Q5**
Logic, Set 53: Rearranging letters · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same letters, each as many times, in a different order. Spaces do not count. Which pair of list words, written together, is an anagram of the phrase PAST TEEN?

**A.** STAR PEN
**B.** NEAT POTS
**C.** TASTE PEN
**D.** PETAL NET

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-F2-Q1**
Logic, Set 53: Rearranging letters · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. Every word on the list that is also a list word when read backward has 3 or 4 letters.

**A.** True
**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-F2-Q2**
Logic, Set 53: Rearranging letters · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. Give a word on the list that is also a word on the list when read backward.

Word = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-F2-Q3**
Logic, Set 53: Rearranging letters · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. How many words on the list can be spelled using the letters of STONE, each letter at most once?

Words: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-F2-Q4**
Logic, Set 53: Rearranging letters · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same letters, each as many times, in a different order. How many words on the list use exactly the letters of NOTES, counting NOTES itself?

Words: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S53-F2-Q5**
Logic, Set 53: Rearranging letters · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.WORD.03 (local) · MP7, MP1

# Set 54: Weighings and the least information needed

Local outcome LOGIC.SEARCH.02: Plan weighings or questions and find the least number that always identifies the answer

Puzzle set · genre: balance puzzles · MP1, MP3

MAIN: Plan weighings or questions and find the least number that always identifies the answer
BACKWARD 1: Strategy: each weighing has three outcomes
FORWARD 1: ADVANCED APPLICATION: Bigger groups and comparisons
FORWARD 2: EXPERT CHALLENGE: Least weighings, proved

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 9 coins and one is heavier. What is the least number of weighings that always finds the heavy coin?

Weighings: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-M-Q1**
Logic, Set 54: Weighings and the least information needed · MAIN · Q1/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 9 coins and one is heavier. Which first weighing lets you always finish in the fewest total weighings?

**A.** 3 coins against 3 coins
**B.** 1 coin against 1 coin
**C.** 2 coins against 2 coins
**D.** 4 coins against 4 coins

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-M-Q2**
Logic, Set 54: Weighings and the least information needed · MAIN · Q2/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. With 3 weighings you can always find the heavier coin among 27 coins.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-M-Q3**
Logic, Set 54: Weighings and the least information needed · MAIN · Q3/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. What is the largest number of coins n for which 2 weighings always find the one heavier coin?

n = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-M-Q4**
Logic, Set 54: Weighings and the least information needed · MAIN · Q4/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 10 coins and one is heavier. What is the least number of weighings that always finds the heavy coin?

Weighings: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-M-Q5**
Logic, Set 54: Weighings and the least information needed · MAIN · Q5/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

How many different outcomes can one weighing on a balance scale have?

Outcomes: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-B1-Q1**
Logic, Set 54: Weighings and the least information needed · BACKWARD 1 (guided strategy) · Q1/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. With 3 coins and one heavier, one weighing always finds the heavy coin.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-B1-Q2**
Logic, Set 54: Weighings and the least information needed · BACKWARD 1 (guided strategy) · Q2/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 5 coins and one is heavier. You weigh 2 against 2 and they balance. What do you know?

**A.** Nothing yet.
**B.** The fifth coin is the heavy one.
**C.** You must weigh again to know.
**D.** One of the four weighed coins is heavy.

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S54-B1-Q3**
Logic, Set 54: Weighings and the least information needed · BACKWARD 1 (guided strategy) · Q3/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

How many outcome patterns do 2 weighings have in all?

Patterns: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S54-B1-Q4**
Logic, Set 54: Weighings and the least information needed · BACKWARD 1 (guided strategy) · Q4/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. With 4 coins and one heavier, one weighing always finds the heavy coin.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-B1-Q5**
Logic, Set 54: Weighings and the least information needed · BACKWARD 1 (guided strategy) · Q5/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. You may use 4 weighings. What is the largest number of coins among which you can always find the one heavier coin?

Coins: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S54-F1-Q1**
Logic, Set 54: Weighings and the least information needed · FORWARD 1: ADVANCED APPLICATION · Q1/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 6 coins and one is heavier. You weigh 2 against 2 and the left side goes down. What should the next weighing be?

**A.** The two coins left off against each other

**B.** All six coins again, 3 against 3

**C.** One left-side coin against a right-side coin

**D.** One left-side coin against the other left-side coin

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-F1-Q2**
Logic, Set 54: Weighings and the least information needed · FORWARD 1: ADVANCED APPLICATION · Q2/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

You have 3 weights of different sizes and compare them two at a time on a balance. Finding the heaviest always takes at least 2 weighings.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-F1-Q3**
Logic, Set 54: Weighings and the least information needed · FORWARD 1: ADVANCED APPLICATION · Q3/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

You have 8 weights of different sizes and compare them two at a time on a balance. What is the least number of weighings that always finds the heaviest?

Weighings: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S54-F1-Q4**
Logic, Set 54: Weighings and the least information needed · FORWARD 1: ADVANCED APPLICATION · Q4/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. Give a number of coins n from 20 to 30 for which 3 weighings always find the one heavier coin.

n = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S54-F1-Q5**
Logic, Set 54: Weighings and the least information needed · FORWARD 1: ADVANCED APPLICATION · Q5/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 7 coins and one is LIGHTER (not heavier). What is the least number of weighings that always finds it?

Weighings: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-F2-Q1**
Logic, Set 54: Weighings and the least information needed · FORWARD 2: EXPERT CHALLENGE · Q1/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. Why can 2 weighings NOT always find the heavy coin among 10?

**A.** Two weighings give only 9 outcome patterns, but there are 10 possible heavy coins.

**B.** You cannot put 5 coins on each side.

**C.** 10 is an even number.

**D.** The first weighing must use all 10 coins.

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-F2-Q2**
Logic, Set 54: Weighings and the least information needed · FORWARD 2: EXPERT CHALLENGE · Q2/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. With 4 coins and one heavier, 2 weighings always find the heavy coin.

**A.** True

**B.** False

**REFERENCE LABEL: NOT PART OF THE QUESTION ID S54-F2-Q3**
Logic, Set 54: Weighings and the least information needed · FORWARD 2: EXPERT CHALLENGE · Q3/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. Give a number of coins n from 20 to 90 for which 3 weighings are NOT enough but 4 weighings are enough to always find the one heavier coin.

n = _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S54-F2-Q4**
Logic, Set 54: Weighings and the least information needed · FORWARD 2: EXPERT CHALLENGE · Q4/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3

You have 4 weights of different sizes and compare them two at a time on a balance. What is the least number of weighings that always finds both the heaviest and the lightest?

Weighings: _____

**REFERENCE LABEL: NOT PART OF THE QUESTION  ID S54-F2-Q5**
Logic, Set 54: Weighings and the least information needed · FORWARD 2: EXPERT CHALLENGE · Q5/5 · Outcome LOGIC.SEARCH.02 (local) · MP1, MP3